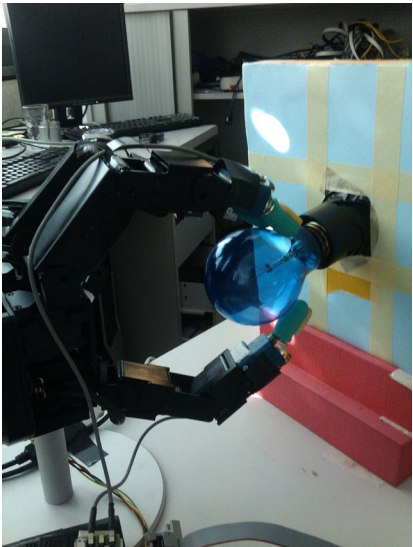
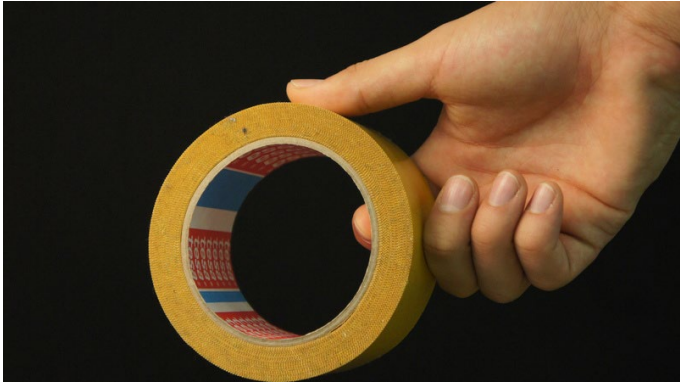




Robotic Grasping and Manipulation

- Grasp Planning

Miao Li



2023, Wuhan University

WeChat: 15527576906

Email: limiao712@gmail.com



How to grasp an object / objects with a robotic hand?

- Background

Catching an empty bottle
(KUKA LWR 4+ and Allegro Hand)



Grand Challenge:



thing and um rod Brooks we may all know
said

“The ability to grasp
arbitrary objects...would
have significant impact in
factories, warehouses,
and homes.”

ROD BROOKS, FEBRUARY 2017



AMAZON.COM ANNOUNCES FIRST QUARTER RESULTS

SEATTLE—(BUSINESS WIRE) April 27, 2023—Amazon.com, Inc. (NASDAQ: AMZN) today announced financial results for its first quarter ended March 31, 2023.

technology, vehicle safety controls, and engineered ergonomic solutions to reduce risks for employees, partners, and communities.

Reached 1 billion packages handled by Robin, a robotic system used across Amazon's operations network in the U.S., Canada, and Europe. Robin robotic systems use artificial intelligence, computer vision, and machine learning to help employees handle and sort customer packages before shipment. For employees, this reduces the number of repetitive tasks, improving safety and helping them focus their time and energy on activities that make better use of their talents.

- Ranked No. 2 on *Fortune* magazine's World's Most Admired Companies list for the seventh year in a row. The annual ranking is determined by a survey of top executives and analysts, and companies are evaluated on factors including the quality of their management and products, commitments to social responsibility, and ability to attract talent.
- Introduced Leadership Liftoff, an onboarding program for operations area managers. New managers enrolled in the program participate in group sessions, on-the-job training, and self-paced learning to help them gain skills and confidence to lead successful teams and foster connections between new leaders.



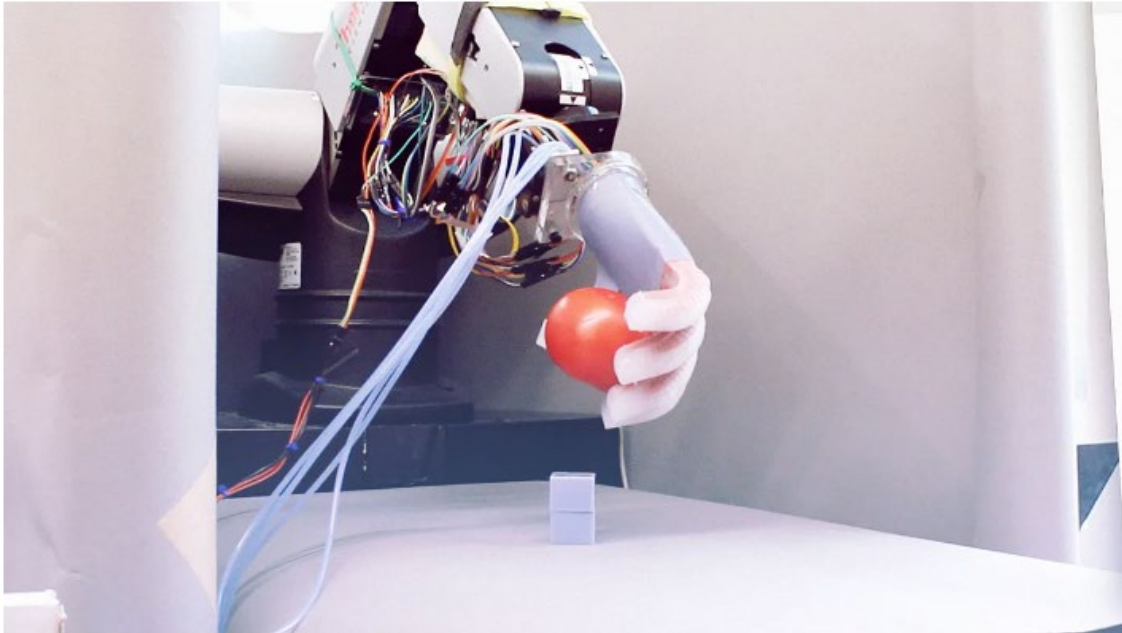
Robin, one of the most complex stationary robot arm systems Amazon has ever built, brings many core technologies to new levels and acts as a glimpse into the possibilities of combining vision, manipulation and machine learning.



01-12-17 | ROBOT REVOLUTION

Why It's So Hard For Robots To Get A Grip

Robots are all thumbs. Engineers are working to improve dexterity so bots can take over housekeeping drudgework—as well as people's jobs.



Cornell's Prosthetic Hand [Photo: Huichan Zhao] [PHOTO: HUICHAN ZHAO]

Berkeley robotics professor Ken Goldberg is turning an empty coffee mug around and around in his hands. “Oh, it’s so complicated for a robot to be able to make sense of that kind of data,” he says, eyeing his fingers grasping the cup with a look of wonder. Artificial intelligence is taking on complex cognitive tasks, such as assisting in legal and medical research, but a manual job like picking up laundry off the floor is still science fiction. It’s a long way from Roomba to Rosie the Robot. Universities like Berkeley and Cornell and companies like Amazon and Toyota are working to close the gap with mechanical hands that approach human dexterity.



Berkeley's YuMi Packing robot [PHOTO: SEAN CAPTAIN]

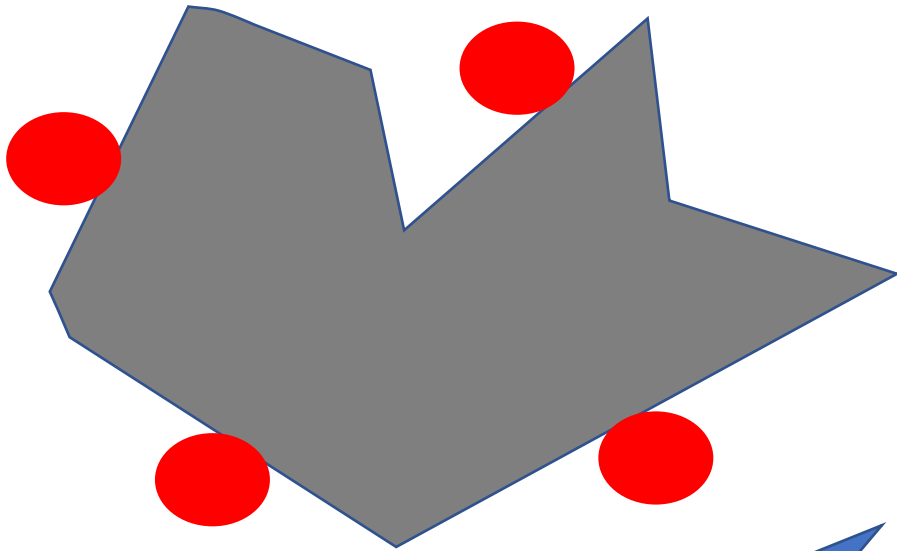
<https://www.fastcompany.com/3066863/why-its-so-hard-for-robots-to-get-a-grip#:~:text=Robots%20have%20to%20overcome%20a,human%20hand%2C%E2%80%9D%20says%20Shepherd.>



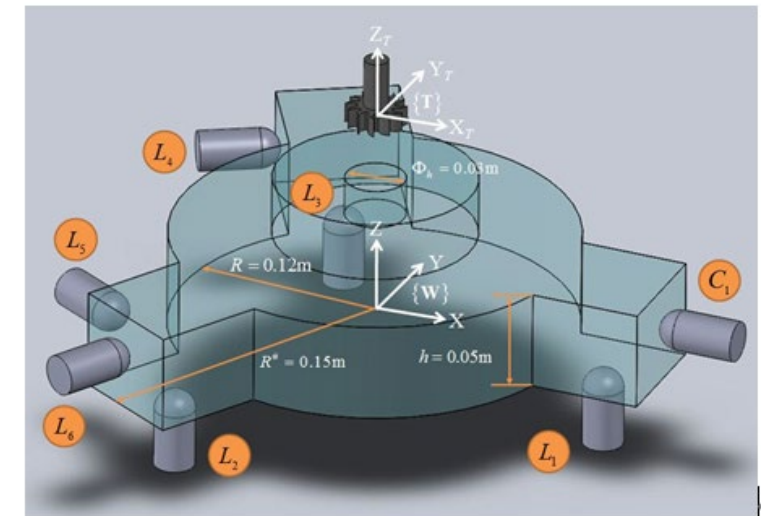
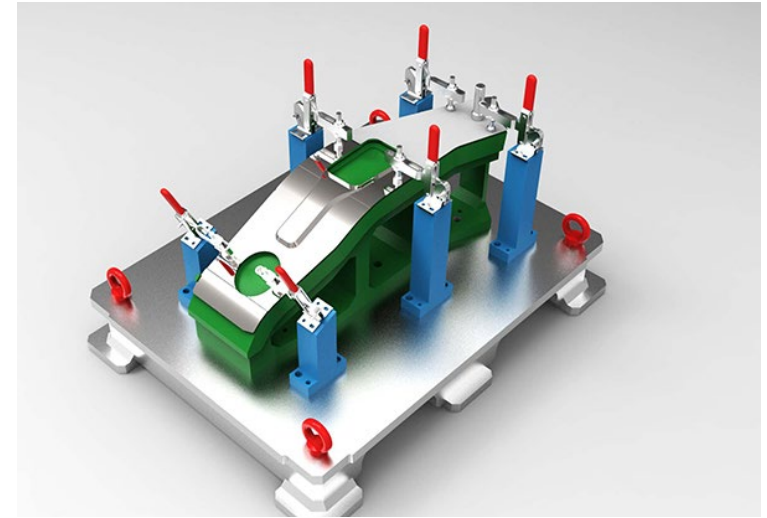




Object-centric approach



How to find the
desired contact points?





Hand-centric approach

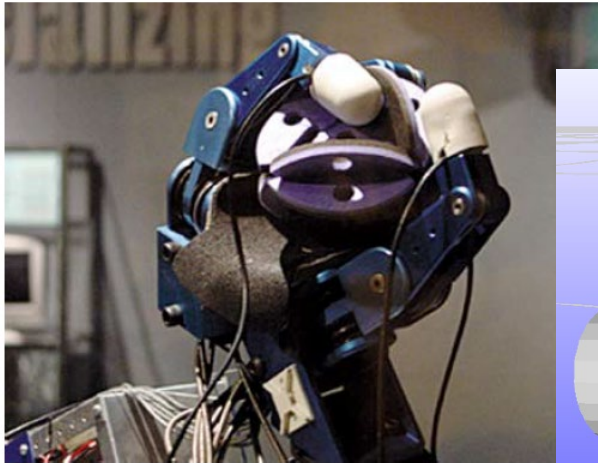


Fig. 28.1 The Salisbury hand grasping an object

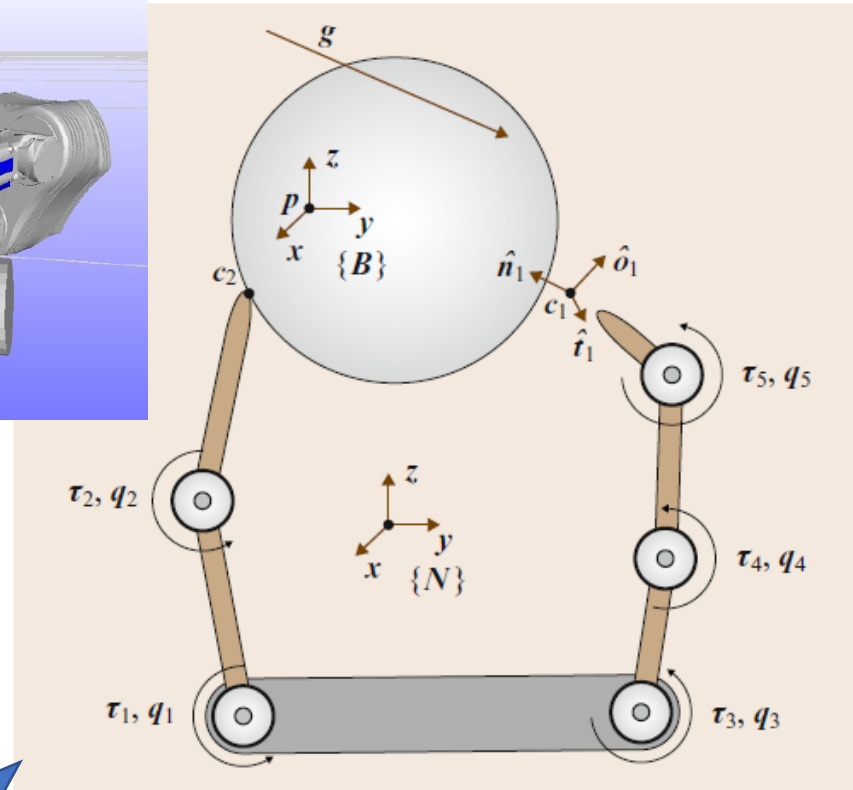
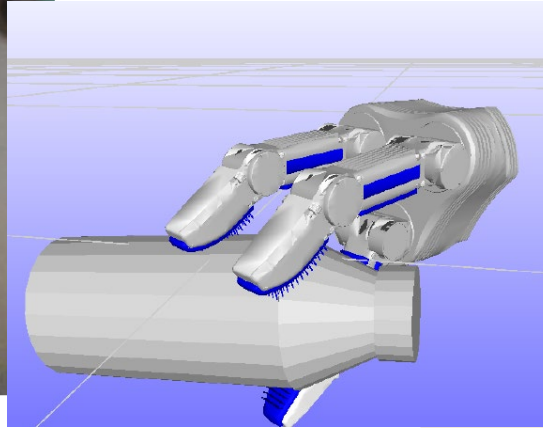


Fig. 28.2 Main quantities for grasp analysis

How to find/design the desired hand configuration?

New hand design,
soft hand, our
project 1



Hand-centric approach



*Annual Review of Control, Robotics, and
Autonomous Systems*

A Century of Robotic Hands

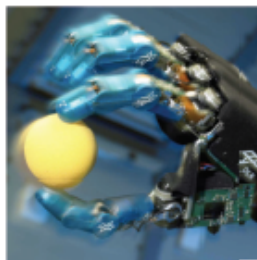
C. Piazza,¹ G. Grioli,² M.G. Catalano,² and A. Bicchi^{1,2}

¹Centro di Ricerca "E. Piaggio" and Dipartimento di Ingegneria Informatica, Università di Pisa,
56122 Pisa, Italy; email: cristina.piazza@ing.unipi.it

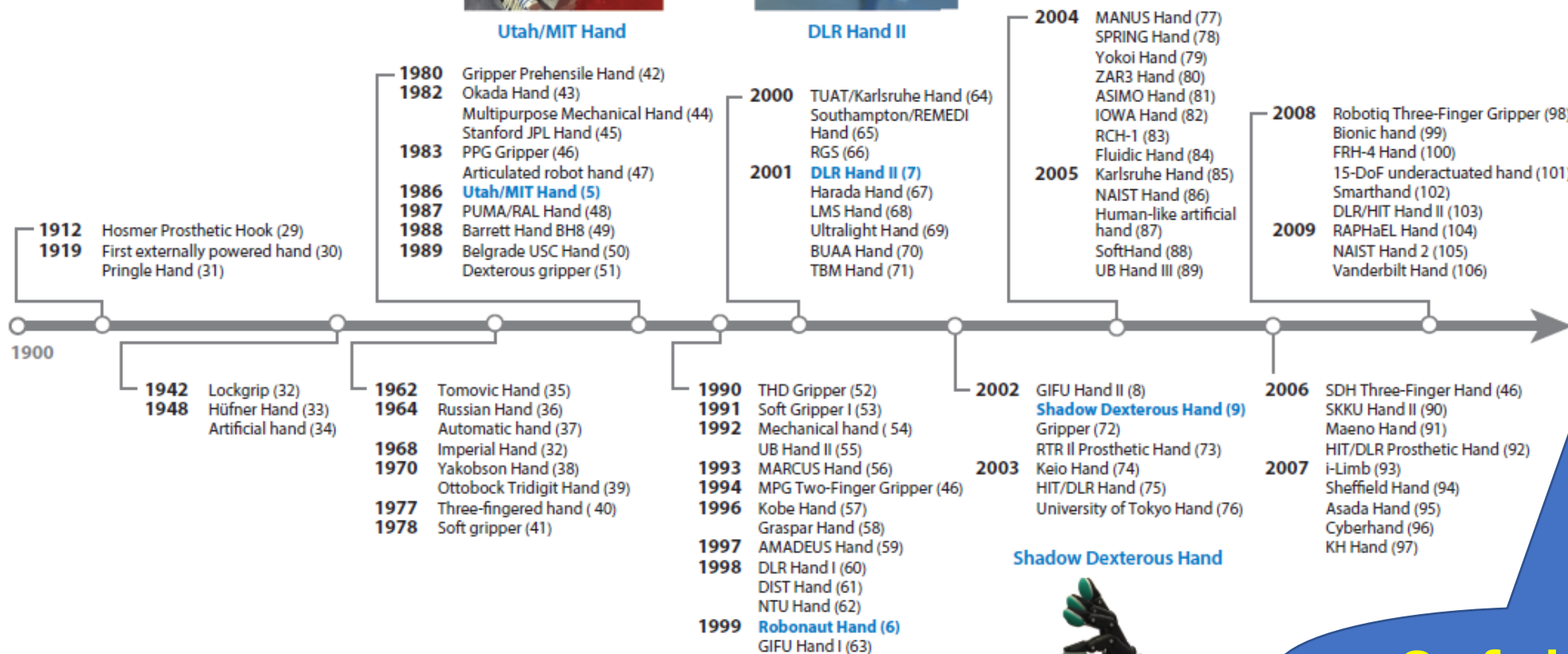
²Soft Robotics for Human Cooperation and Rehabilitation, Istituto Italiano di Tecnologia,
16163 Genova, Italy



Utah/MIT Hand



DLR Hand II



Robonaut Hand



Shadow Dexterous Hand

Soft hand



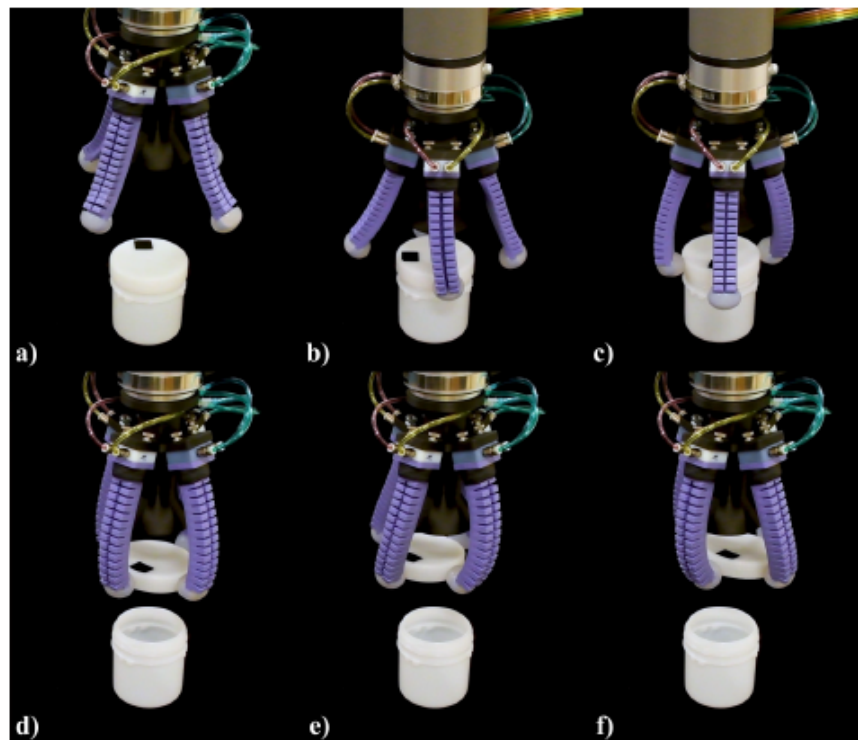
Hand-centric approach

A Dexterous Soft Robotic Hand for Delicate In-Hand Manipulation

Sylvain Abondance^{*1,2}, Clark B. Teeple^{*1}, and Robert J. Wood¹

Abstract—In this work, we show that soft robotic hands provide a robust means of performing basic primitives of in-hand manipulation in the presence of uncertainty. We first discuss the design of a prototype hand with dexterous soft fingers capable of moving objects within the hand using several basic motion primitives. We then empirically validate the ability of the hand to perform the desired object motion primitives while still maintaining strong grasping capabilities. Based on these primitives, we examine a simple, heuristic finger gait which enables continuous object rotation for a wide variety of object shapes and sizes. Finally, we demonstrate the utility of our dexterous soft robotic hand in three real-world cases: unscrewing the cap of a jar, orienting food items for packaging, and gravity compensation during grasping. Overall, we show that even for complex tasks such as in-hand manipulation, soft robots can perform robustly without the need for local sensing or complex control.

Index Terms—In-Hand Manipulation, Soft Robot Applications, Dexterous Manipulation

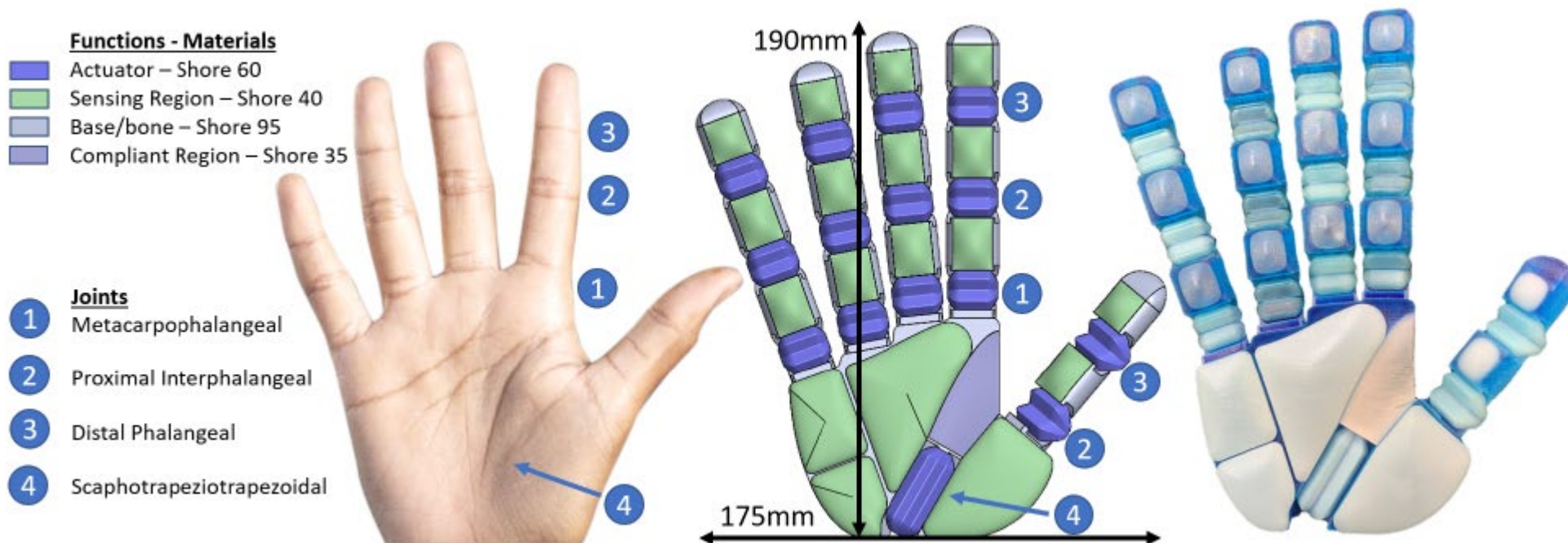




Hand-centric approach

Design of a 3D-Printed Soft Robotic Hand with Integrated Distributed Tactile Sensing

Oliver Shorthose, Alessandro Albini, Liang He, and Perla Maiolino*



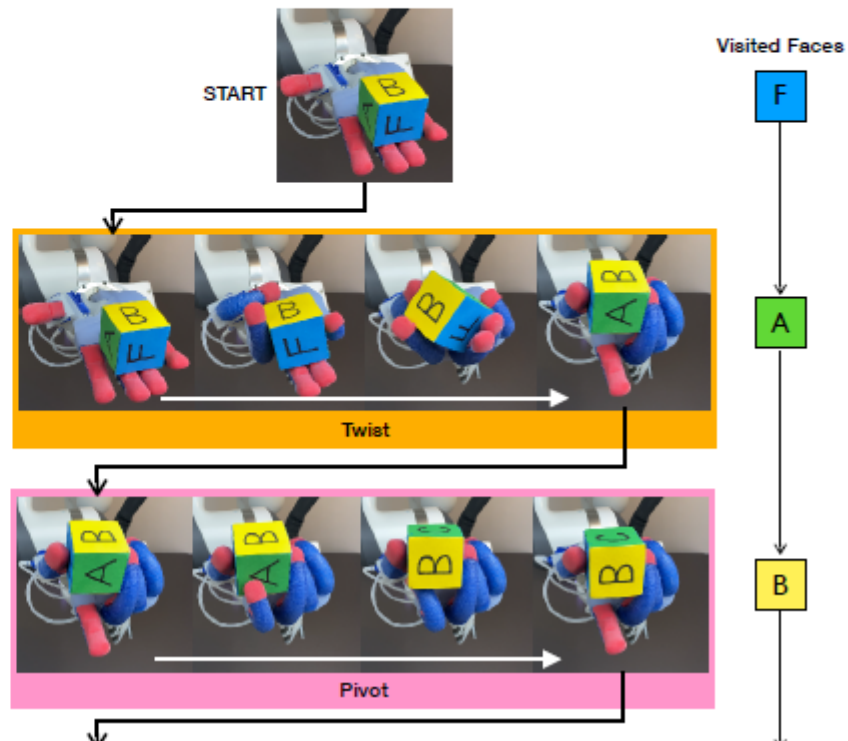


Hand-centric approach

Surprisingly Robust In-Hand Manipulation: An Empirical Study

Aditya Bhatt* Adrian Sieler* Steffen Puhlmann Oliver Brock

Abstract—We present in-hand manipulation skills on a dexterous, compliant, anthropomorphic hand. Even though these skills were derived in a simplistic manner, they exhibit surprising robustness to variations in shape, size, weight, and placement of the manipulated object. They are also very insensitive to variation of execution speeds, ranging from highly dynamic to quasi-static. The robustness of the skills leads to compositional properties that enable extended and robust manipulation programs. To explain the surprising robustness of the in-hand manipulation skills, we performed a detailed, empirical analysis of the skills' performance. From this analysis, we identify three principles for skill design: 1) Exploiting the hardware's innate ability to drive hard-to-model contact dynamics. 2) Taking actions to constrain these interactions, funneling the system into a narrow set of possibilities. 3) Composing such action sequences into complex manipulation programs. We believe that these principles constitute an important foundation for robust robotic in-hand manipulation, and possibly for manipulation in general.





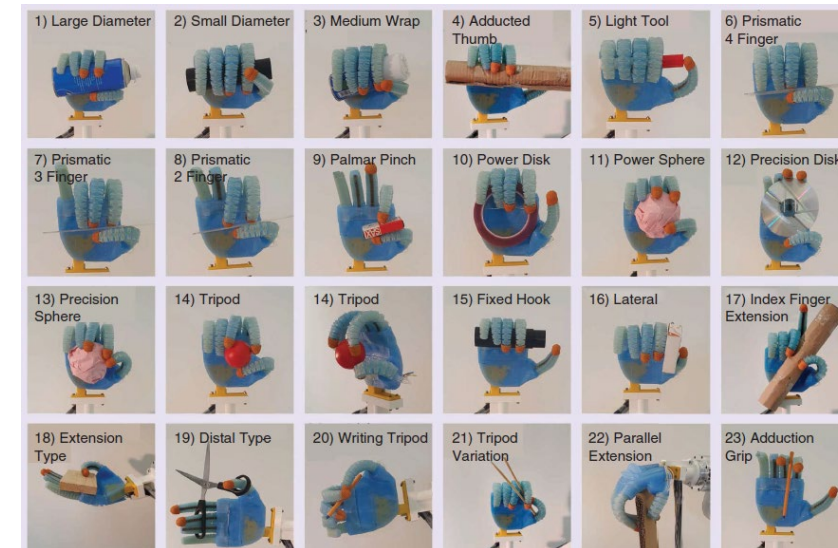
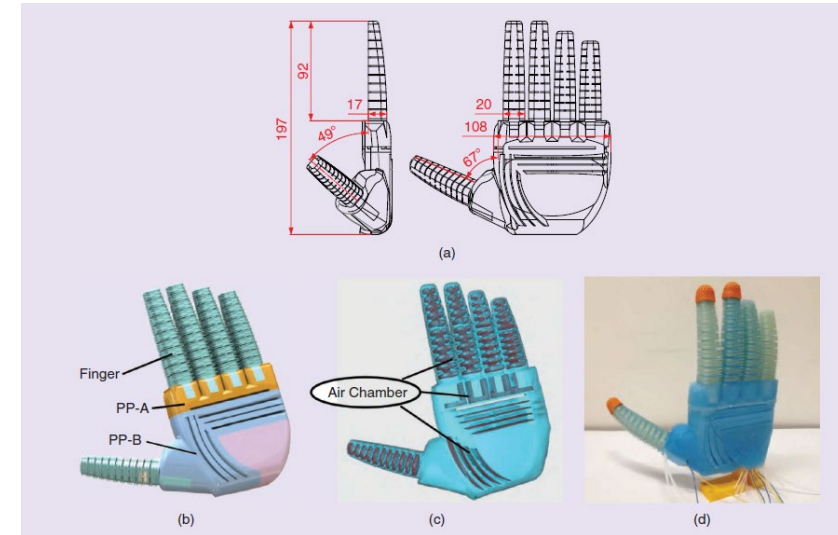
Hand-centric approach



A Novel Soft Robotic Hand Design With Human-Inspired Soft Palm

Achieving a Great Diversity of Grasps

By Haihang Wang, Fares J. Abu-Dakka, Tran Nguyen Le, Ville Kyrki, and He Xu





Hand-centric approach

IEEE ROBOTICS AND AUTOMATION LETTERS. PREPRINT VERSION. JANUARY, 2022

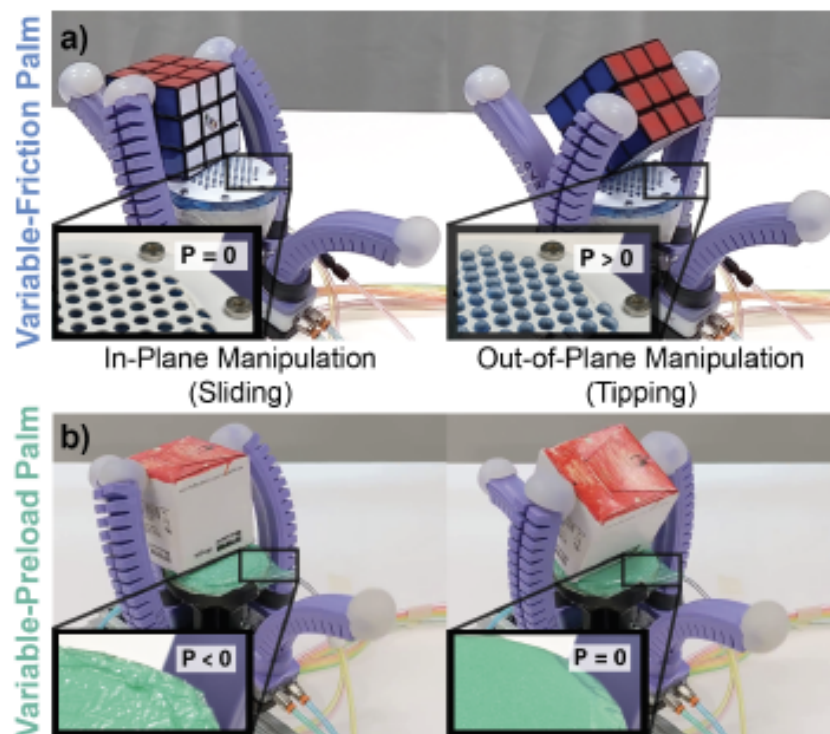
1

Controlling Palm-Object Interactions via Friction for Enhanced In-Hand Manipulation

Clark B. Teeple¹, Buse Aktaş¹, Michelle C. Yuen¹, Grace R. Kim¹, Robert D. Howe^{1,2}, and Robert J. Wood¹

Abstract—In-hand manipulation is necessary, yet challenging for robotic hands as robots begin to interact with real-world objects and environments. The intentional incorporation of an active palm can lead to improved dexterity. In this work, we demonstrate how controlling the palm-object interaction enables access to different sets of motion primitives. We first discuss how various properties of the palm affect in-hand manipulation, as well as grasping. Here, we focus on varying the friction force between the object and the palm by either changing the coefficient of friction or the normal load. We explore how these two design factors can be actively controlled to enable 5-degree-of-freedom object motion within the hand. We present *variable-friction* and *variable-preload* palms that enable on-the-fly switching between in-plane and out-of-plane manipulation modes. Additionally, we discuss their impact on grasp stability. Finally, we demonstrate how an object can be translated and pivoted within the hand by sequentially performing synchronized palm and finger actions.

Index Terms—In-Hand Manipulation, Dexterous Manipulation, Soft Robot Applications, Soft Robot Materials and Design





Hand-centric approach



Article

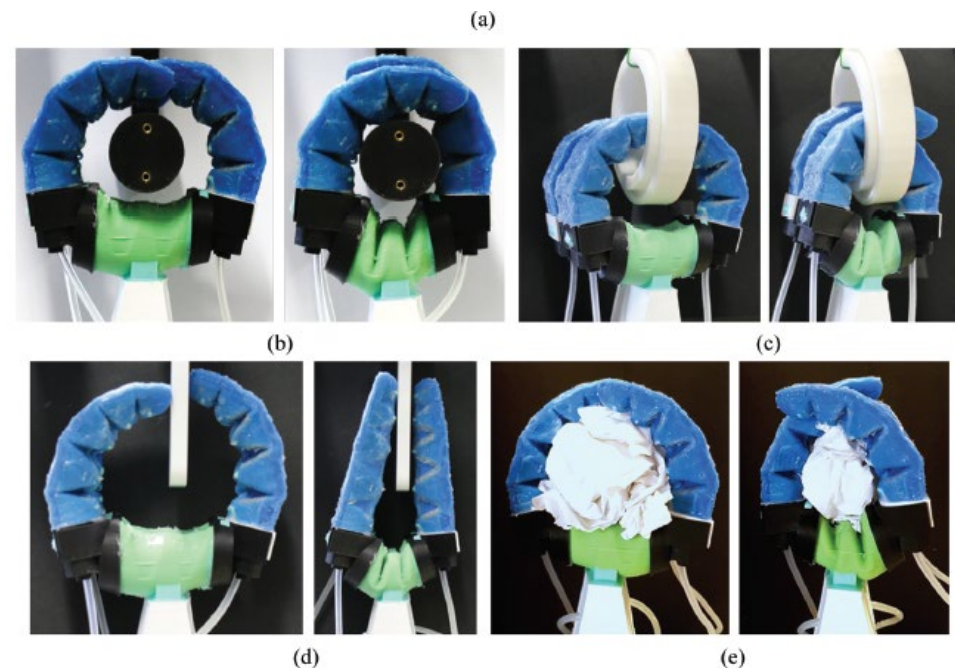
Design and characterization of a hybrid soft gripper with active palm pose control

Vignesh Subramaniam¹ , Snehal Jain², Jai Agarwal³ and Pablo Valdivia y Alvarado^{2,4} 

The International Journal of
Robotics Research
1–18
© The Author(s) 2020
Article reuse guidelines:
sagepub.com/journals-permissions
DOI: 10.1177/0278364920918918
journals.sagepub.com/home/ijr


Abstract

The design and characterization of a soft gripper with an active palm to control grasp postures is presented herein. The gripper structure is a hybrid of soft and stiff components to facilitate integration with traditional arm manipulators. Three fingers and a palm constitute the gripper, all of which are vacuum actuated. Internal wedges are used to tailor the deformation of a soft outer reinforced skin as vacuum collapses the composite structure. A computational finite-element model is proposed to predict finger kinematics. Thanks to its active palm, the gripper is capable of grasping a wide range of part geometries and compliances while achieving a maximum payload of 30 N. The gripper natural softness enables robust open-loop grasping even when components are not properly aligned. Furthermore, the grasp pose of objects with various aspect ratios and compliances can be robustly maintained during manipulation at linear accelerations of up to 15 m/s^2 and angular accelerations of up to 5.23 rad/s^2 .





Hand-centric approach

frontiers
in Robotics and AI

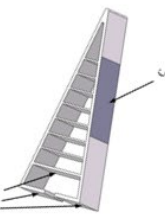
ORIGINAL RESEARCH
published: 10 November 2016
doi: 10.3389/frobt.2016.00070



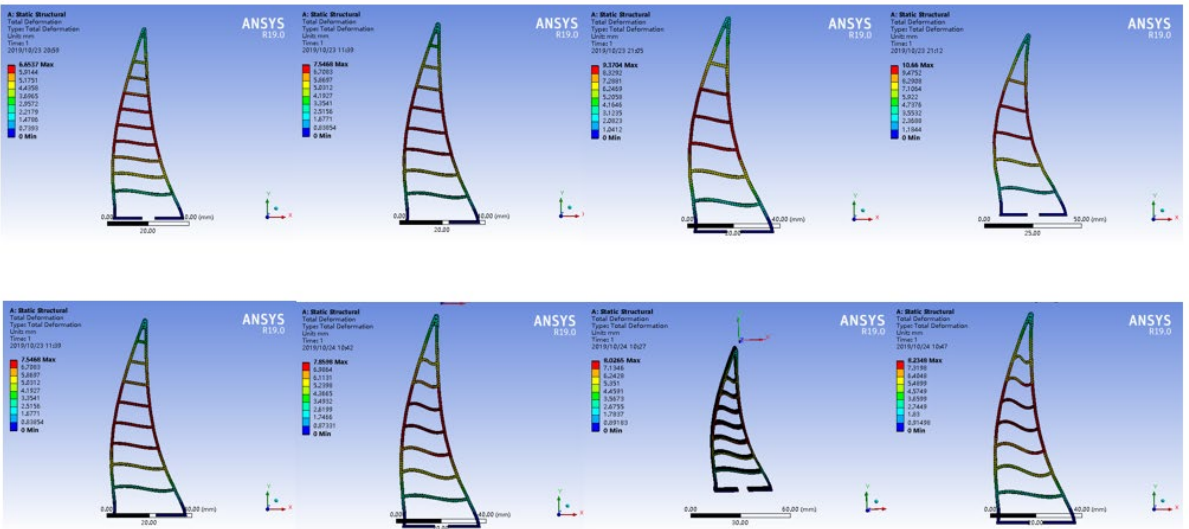
Fin Ray[®] Effect Inspired Soft Robotic Gripper: From the RoboSoft Grand Challenge toward Optimization

Whitney Crooks^{1*}, Gabrielle Vukasin^{1,2}, Maeve O'Sullivan¹, William Messner² and Chris Rogers¹

¹ Department of Mechanical Engineering, Center for Engineering Education and Outreach, Tufts University, Medford, MA, USA, ² Controls, Robotics, Identification, and Signal Processing Laboratory, Department of Mechanical Engineering, Tufts University, Medford, MA, USA



$n = 8\text{mm}, 10\text{mm}, 12\text{mm}, 14\text{mm}$



$a = 0.4\text{mm}, 0.6\text{mm}, 0.8\text{mm}, 1\text{mm}$

frontiers
in Robotics and AI

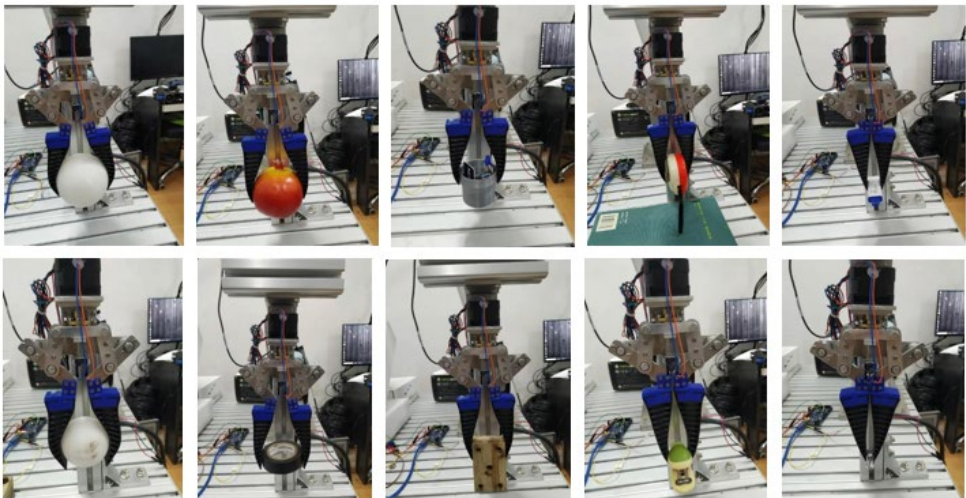
ORIGINAL RESEARCH
published: 12 February 2021
doi: 10.3389/frobt.2020.590076



Learning Optimal Fin-Ray Finger Design for Soft Grasping

Zhifeng Deng¹ and Miao Li^{1,2*}

¹ Learning Algorithms and Soft Manipulation Laboratory, The Institute of Technological Science, School of Power and Mechanical Engineering, Wuhan University, Wuhan, China, ² Wuhan Cobot Technology, Wuhan, China





Hand-centric approach

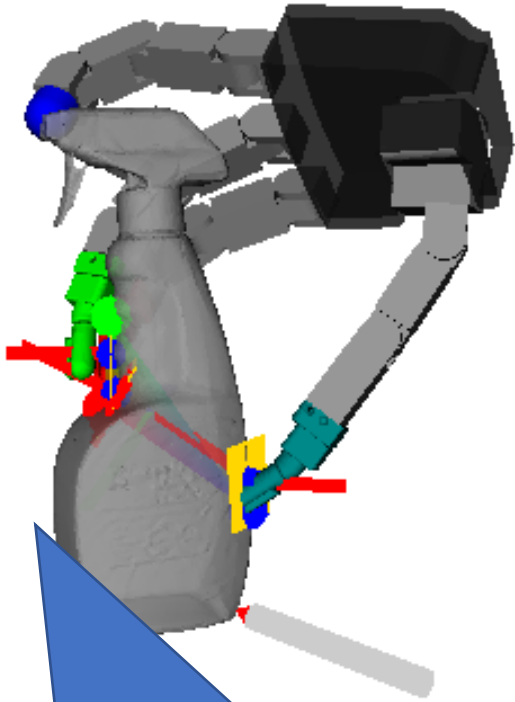
Modeling and Analysis of Soft Robotic Fingers using the Fin Ray Effect

Xiaowei Shan and Lionel Birglen
Polytechnique Montréal, 2019





Hand-centric vs. Object-centric



How to find/design the desired hand configuration?



How to find grasp at contact level?



A conceptual formulation

Objective function

Object representation

Hand representation

Hand-object interaction

Task constraints

What is good for
the grasp?



A conceptual formulation

Objective function

Object representation

Hand representation

Hand-object interaction

Task constraints

Geometry, PCL,
Mesh, 2d
images,...



Object-centric approach -- Object Representation

Grasp Planning as Optimization

$\arg \min_{\theta, H, p, n}$	$f(\theta, H, p, n)$	objective function
subject to:	$h_i(\theta_i, H) - p_i = 0$ $l_i(\theta_i, H) - n_i = 0$	hand constraints
	$g(p_i) = 0$ $\nabla g(p_i) \times n_i = 0$ $\nabla g(p_i) \cdot n_i < 0$	object constraints
	$Q_{\text{task}}(\theta, H, p, n) \in \mathcal{G}_{\text{task}}$	task constraints

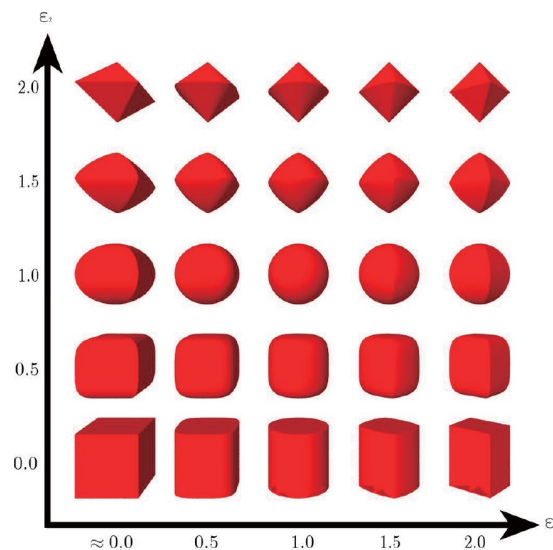
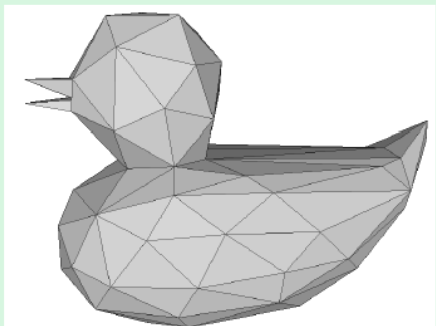


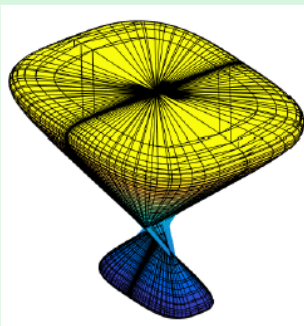
Fig. 1. Different variations of superquadrics



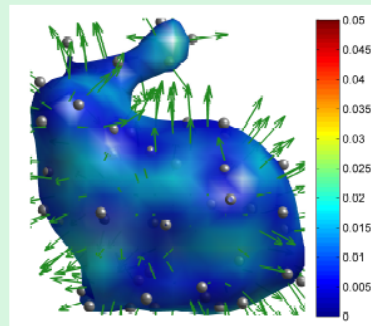
Point clouds



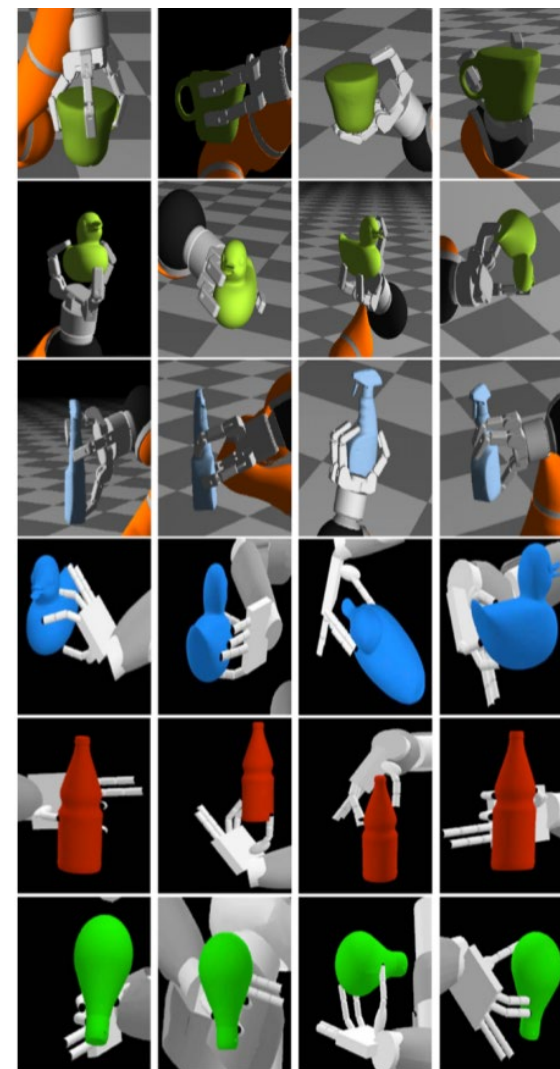
Polyhedron



Superquadrics



GPIS





A conceptual formulation

Objective function

Object representation

Hand representation

Hand-object interaction

Task constraints

Fingertip grasp,
power grasp, eigen
grasp, synergy,
posture, etc ...



A conceptual formulation

Objective function
Object representation
Hand representation
Hand-object interaction
Task constraints

**Contact types,
forces, deformation,
etc ...**



A conceptual formulation

Objective function
Object representation
Hand representation
Hand-object interaction
Task constraints

Affordance, hand-
over, semantics,
chatGPT, ...
(High-level)



Typical Contact Types

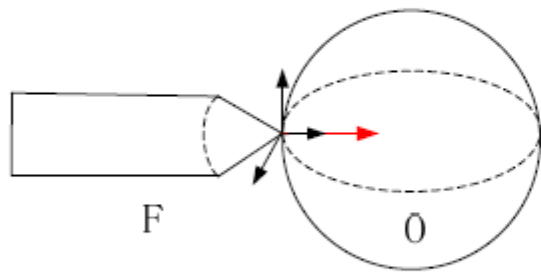


图 2.3 无摩擦点接触

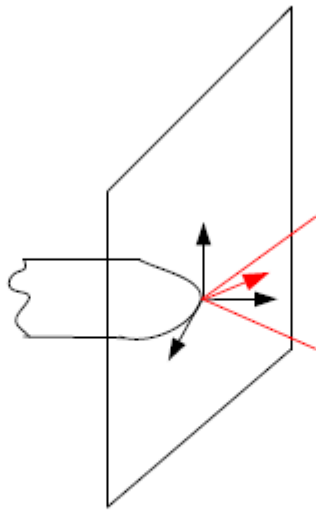


图 2.4 带摩擦点接触

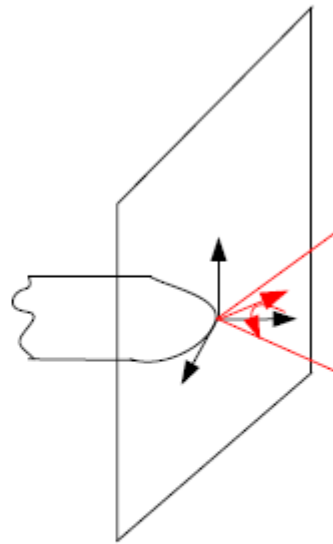


图 2.5 软手指接触



Object-centric approach

分类号 _____

学号 M200870318

学校代码 10487

密级 _____

华中科技大学

硕士学位论文

抓取力分配与不确定性分析

学位申请人：

李森

学科专业：

机械电子工程

指导老师：

杨文玉 教授

答辩日期：

2011 年 1 月 10 日

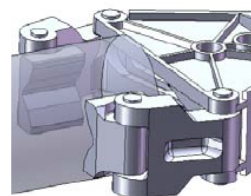


图 5.4 钳口结构

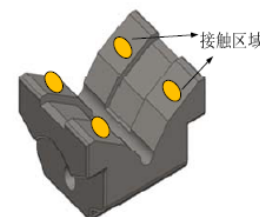


图 5.5 钳口接触区域

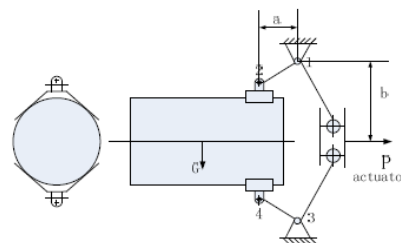


图 5.6 操作机结构简图

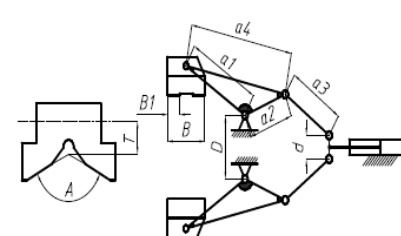


图 5.7 操作机结构参数



Object-centric approach

Dynamic Grasp Adaptation -- From Humans To Robots

THÈSE N° 6908 (2016)

PRÉSENTÉE LE 8 FÉVRIER 2016

À LA FACULTÉ DES SCIENCES ET TECHNIQUES DE L'INGÉNIEUR
LABORATOIRE D'ALGORITHMES ET SYSTÈMES D'APPRENTISSAGE
PROGRAMME DOCTORAL EN SYSTÈMES DE PRODUCTION ET ROBOTIQUE

ÉCOLE POLYTECHNIQUE FÉDÉRALE DE LAUSANNE

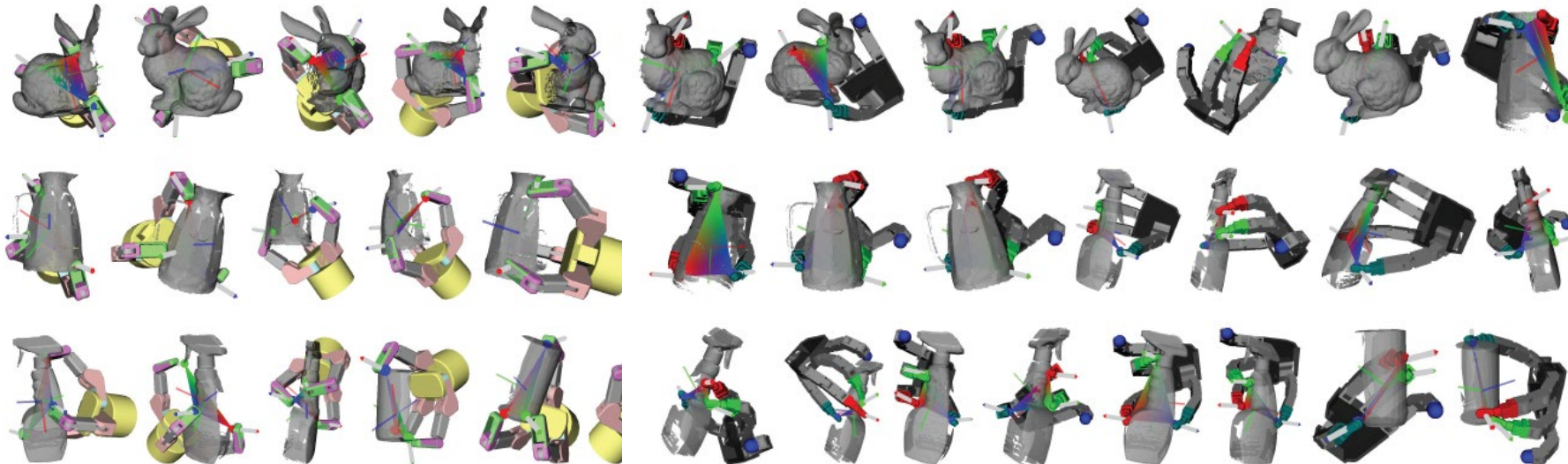
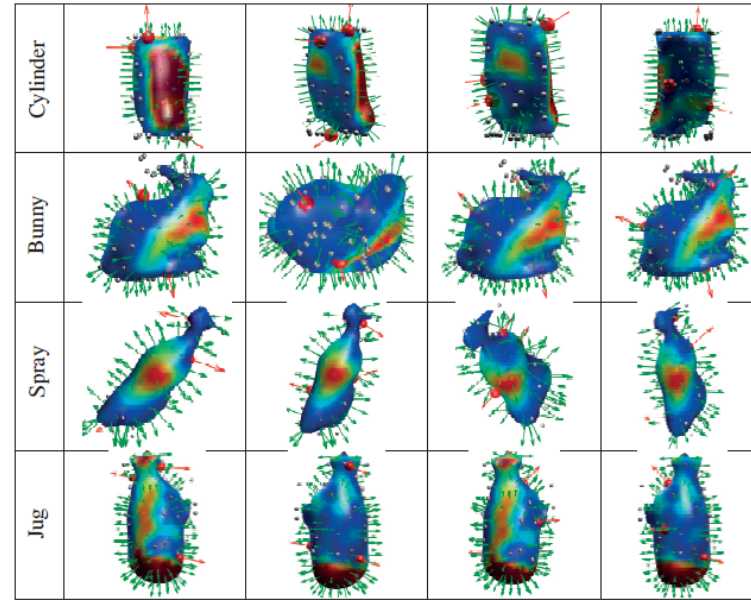
POUR L'OBTENTION DU GRADE DE DOCTEUR ÈS SCIENCES

PAR

Miao LI

acceptée sur proposition du jury:

Prof. J. Paik, présidente du jury
Prof. A. Billard, directrice de thèse
Prof. K. Tahara, rapporteur
Prof. A. Rodriguez, rapporteur
Prof. J. del R. Millan, rapporteur



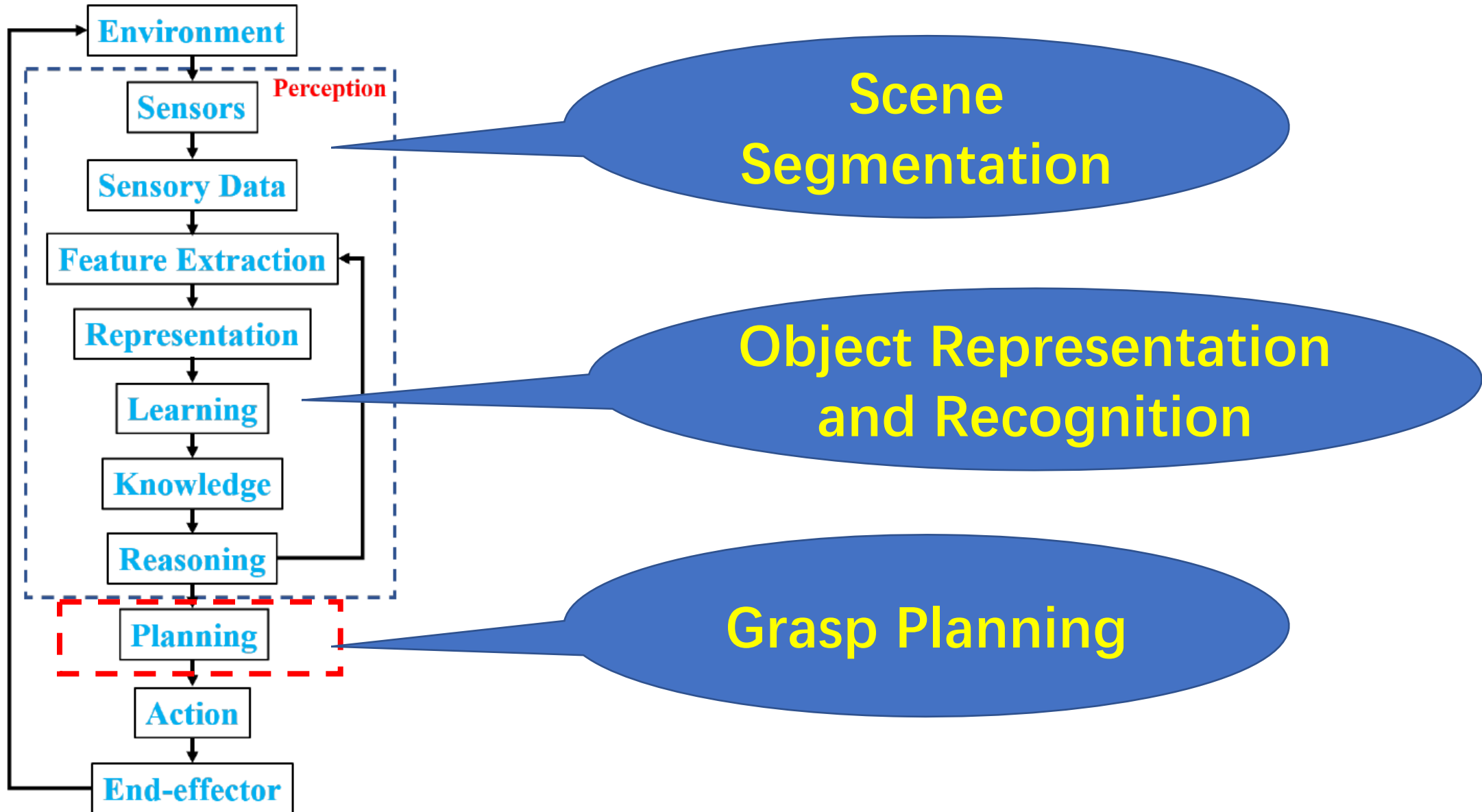


**How to grasp an object / objects with a
robotic hand?**

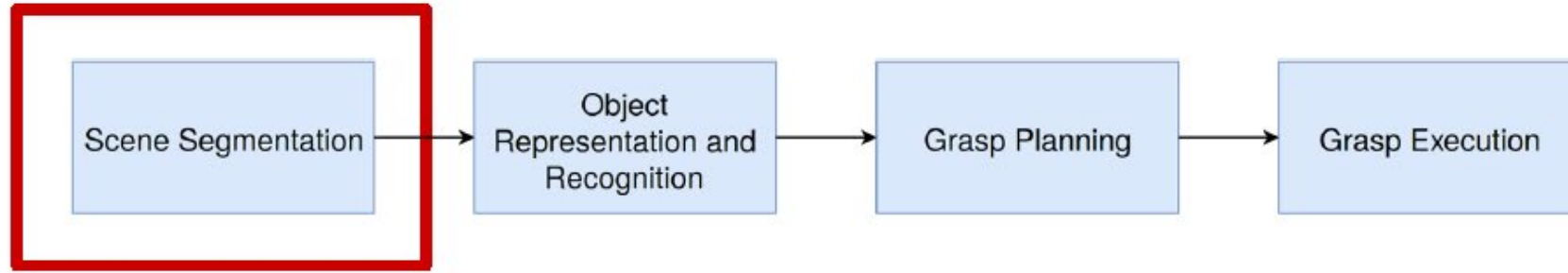
- (2000-2016)



A typical pipeline of robotic grasping



Scene Segmentation



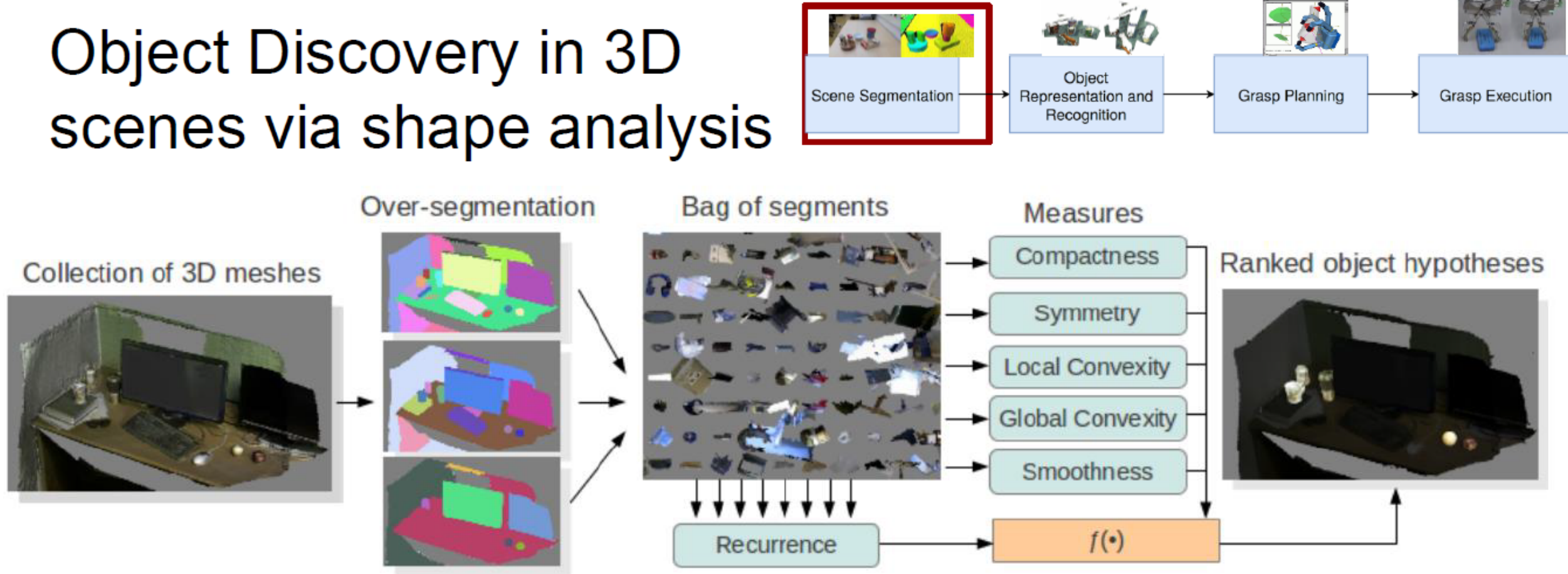
Need to understand what we are going to interact with...

- Euclidean Clustering
- Object Detector



Image From:
Andrej Karpathy, et al. Object discovery in 3d scenes via
shape analysis. ICRA, 2013

Object Discovery in 3D scenes via shape analysis



- Segment 58 scenes using several thresholds
- Train an SVM on 6 handcrafted features to predict whether each segment is an **object** or not?

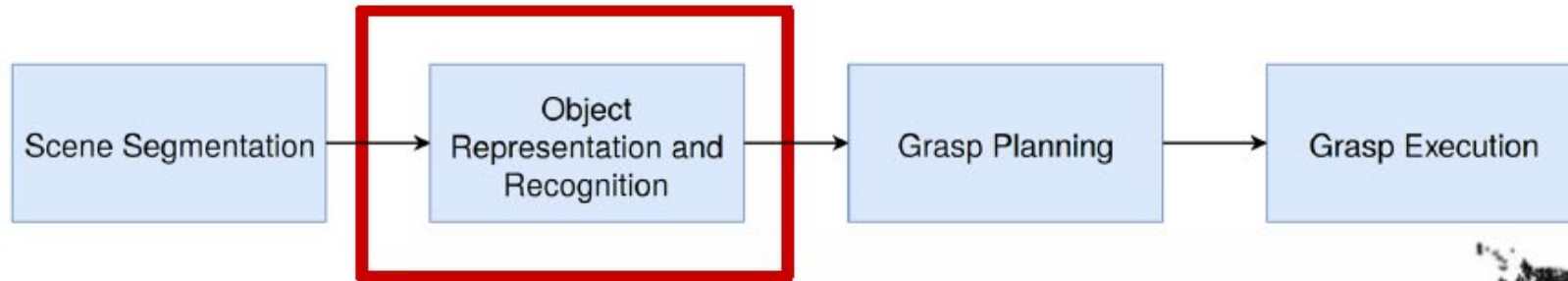
Pros:

- Easy to understand
- Fast

Cons:

- Objectness is vague
- Not dense
- Handcrafted features

Object Modeling



We have a segmented scene, and a region of interest, but the back half is missing....

- General Completion
- Instance Recognition



Image from:

Jeannette Bohg, Matthew Johnson-Roberson, Beatriz Leon, Javier Felip, Xavi Gratal, Niklas Bergström, Danica Kragic, and Antonio Morales. Mind the gap-robotic grasping under incomplete observation. In Robotics and Automation (ICRA), IEEE International Conference on, 2011

Exploiting Symmetries and extrusions for grasping household objects

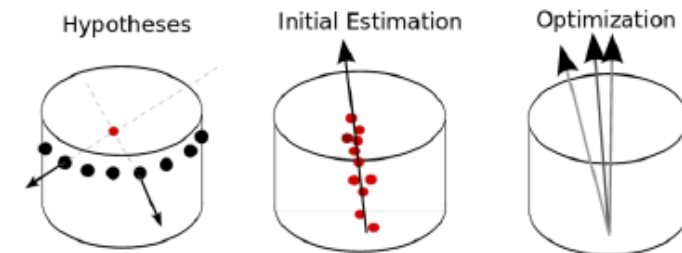
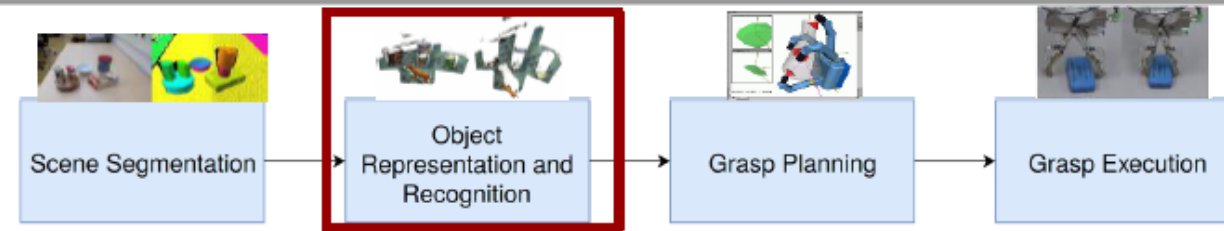
- Reflect points over symmetry plane
- Determine best linear or revolute extrusion for mirrored points.

Pros:

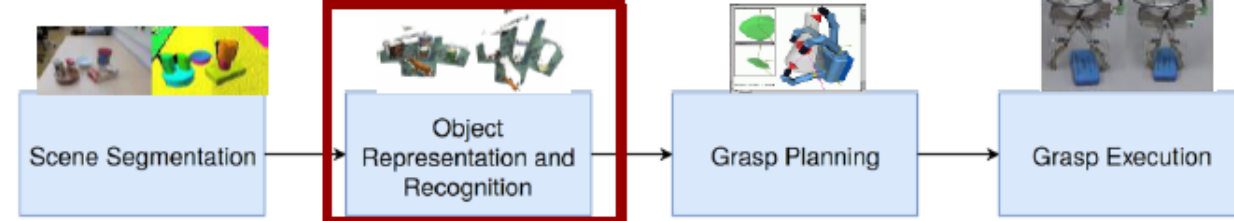
- Many objects exhibit symmetry

Cons:

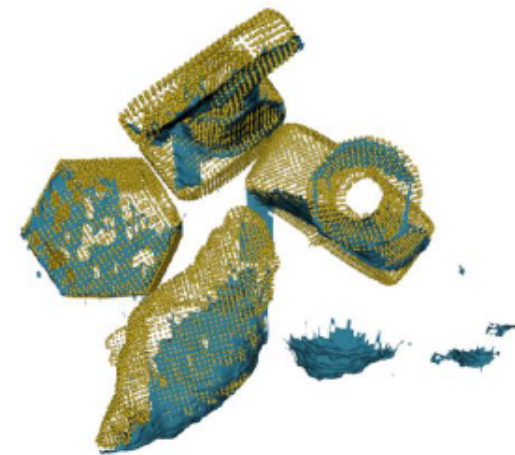
- Just a heuristic



An efficient ransac for 3d object recognition in noisy and occluded scenes



- Database of Object Instances
- Find them in the scene
- Hashtable: pairs of oriented points to model pose.
- Randomly sample hypothesis
- Filter based on evidence and agreement with visible scene



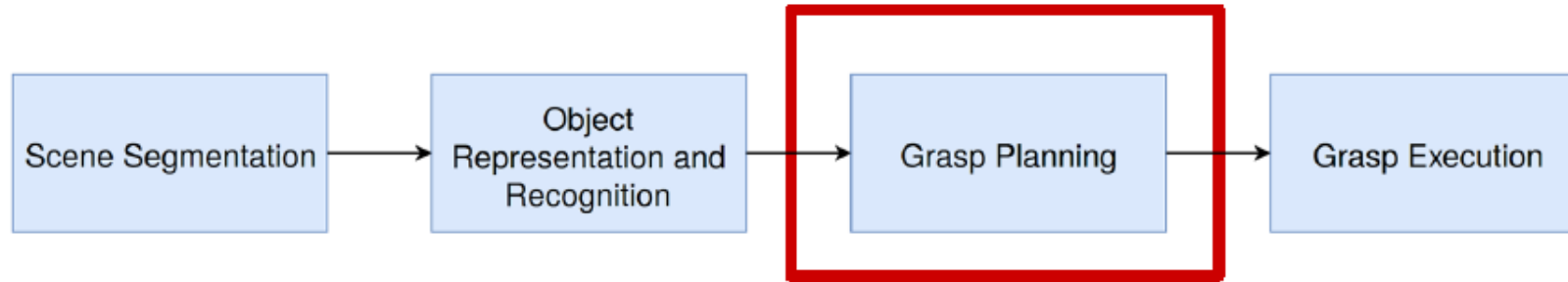
Pros:

- Fast cuda implementation

Cons:

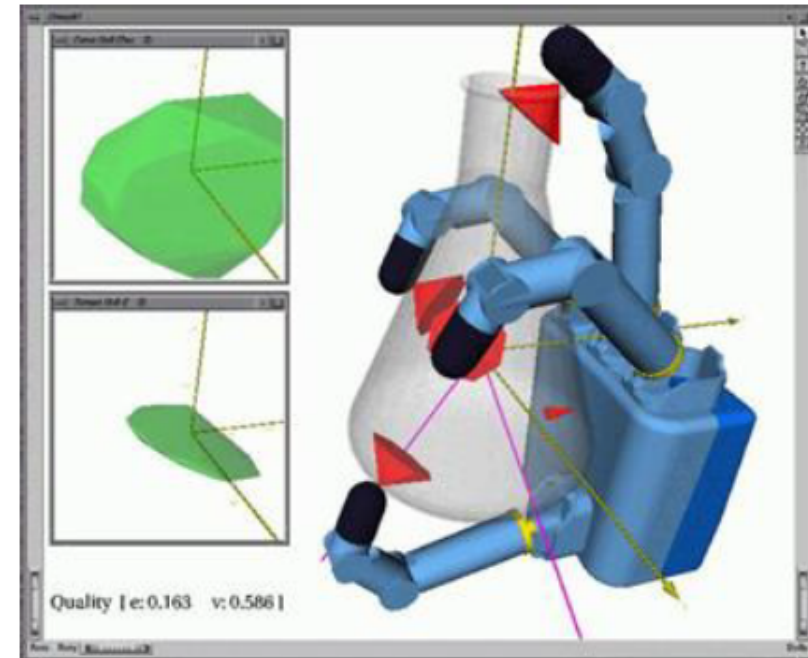
- Exact model matching
- Lots of magic numbers
- Tens of objects only

Grasp Planning

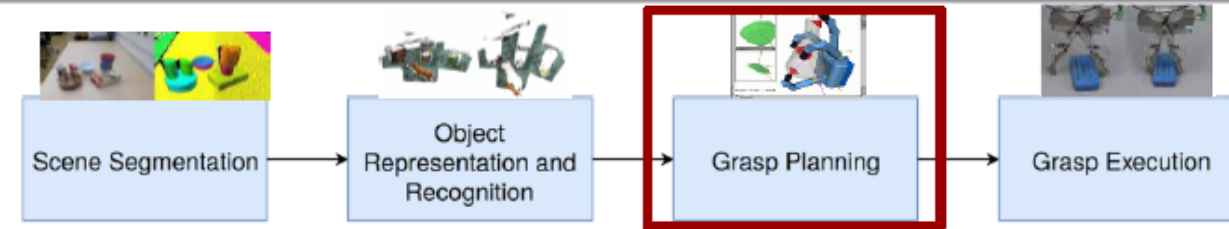


We have a segmented scene, and a completed object to grasp, but how should we pick it up...

- Search for a Grasp
- Precompute a database of grasps
- Grasping rectangles for simple grippers



Hand Posture subspaces for dexterous robotic grasping



- Eigengrasps: First two principal components account for more than 80% of the variance
- Search for “Good” grasps in Eigengrasp space

Pros:

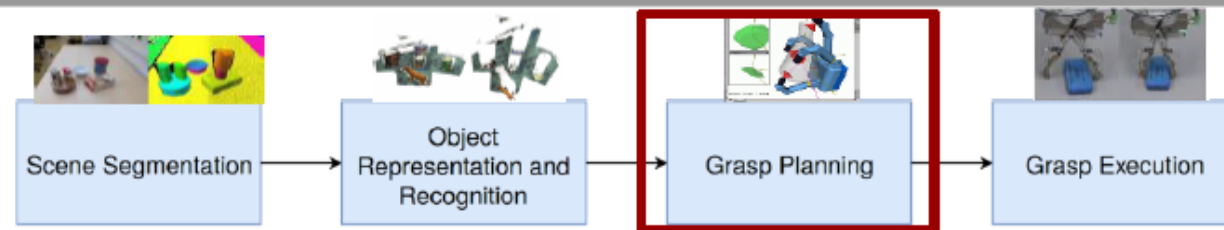
- Reduced dimensionality allows for fast search

Cons:

- Heuristic energy functions
 - Volume Energy

Model	DOFs	Description	Eigengrasp 1		Description	Eigengrasp 2	
			min	max		min	max
Gripper	4	Prox. joints flexion			Dist. joints flexion		
Barrett	4	Spread angle opening			Finger flexion		
DLR	12	Prox. joints flexion Finger abduction			Dist. joints flexion Thumb flexion		
Robonaut	14	Thumb flexion MCP flexion Index abduction			Thumb flexion MCP extension PIP flexion		
Human	20	Thumb rotation Thumb flexion MCP flexion Index abduction			Thumb flexion MCP extension PIP flexion		

Data-driven grasping



Step 1: Creating a grasp database of 3D models annotated with precomputed grasps and quality scores.

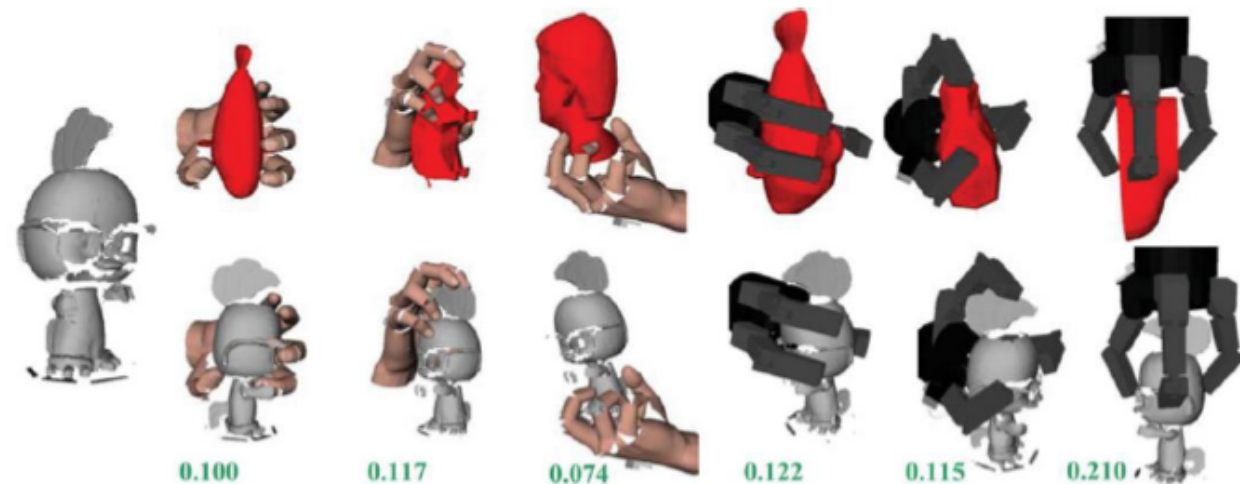
Step 2: Indexing the database for retrieval using partial 3D geometry.

Step 3: Finding matches in the database using only the sensor data, which is typically incomplete.

Step 4: Aligning the object to each of the matched models from the database.

Step 5: Selecting a grasp from the candidate grasps provided by the aligned matches

Step 6: Executing the grasp and evaluating the results.



Pros:

- Data Driven, not a heuristic

Cons:

- Grasp transfer is rigid

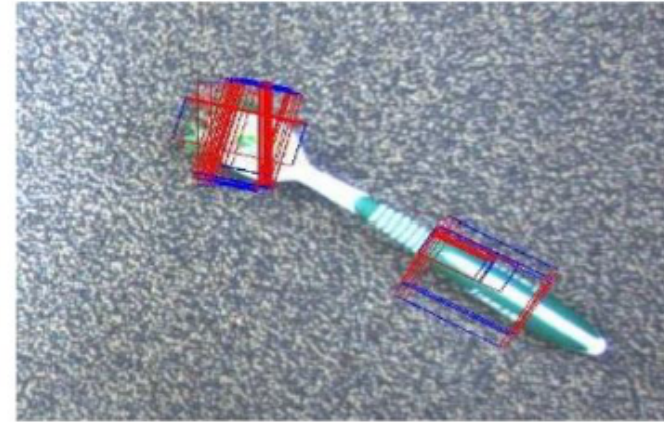
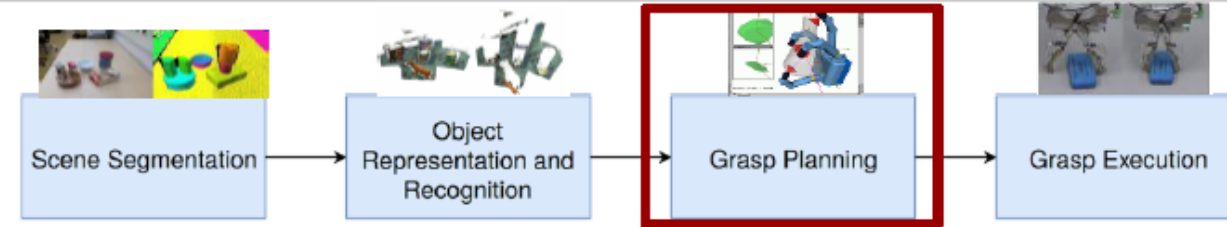
Efficient Grasping from RGBD Images: Learning using a new rectangle representation

Gripper pose is 5 DOF

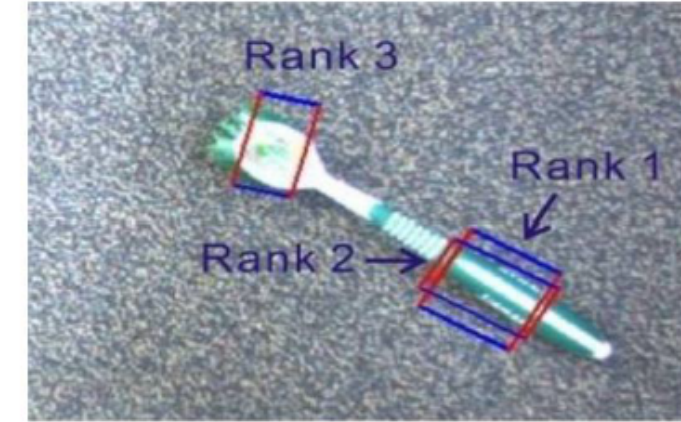
- x, y , width, height, θ

Search:

- Quick first pass search for candidates
- More advanced features to rank candidates



(a) The top 100 rectangles after the first-step search.



(b) The top 3 rectangles after the second-step rank.

Fig. 6: Example results from two-step search.

Pros:

- Data Driven

Cons:

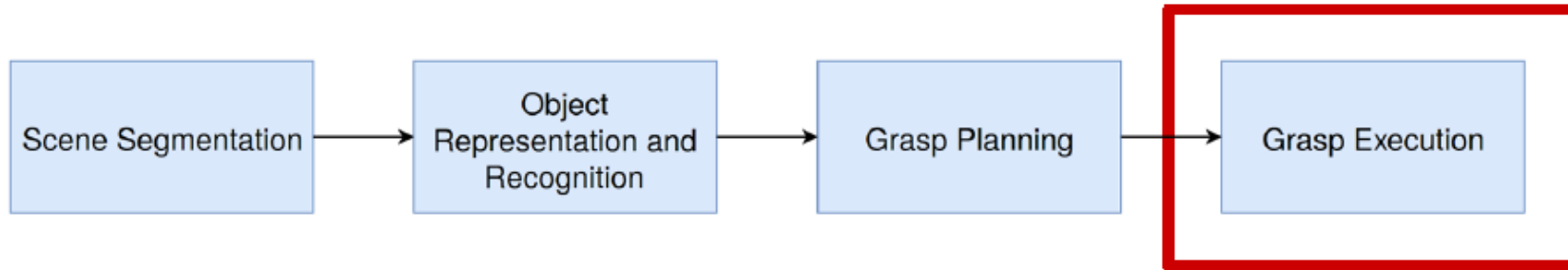
- All grasps are from above

Yun Jiang, Stephen Moseson, and Ashutosh Saxena.

Efficient grasping from rgbd images: Learning using a new rectangle representation.

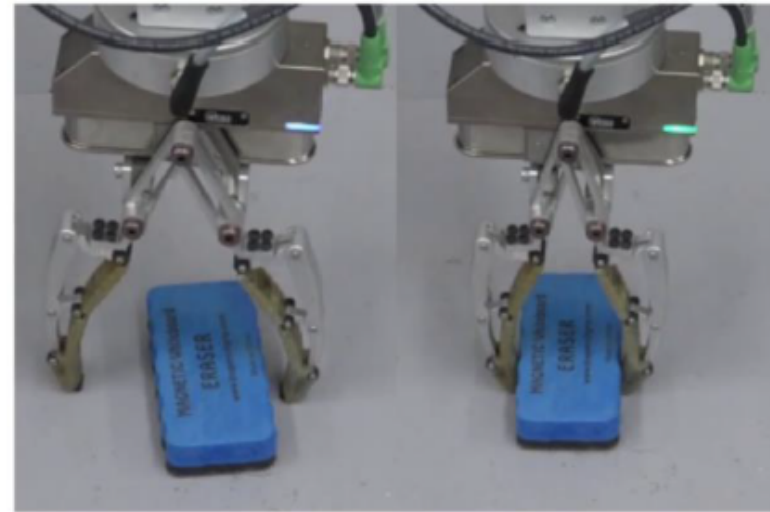
In Robotics and Automation (ICRA), 2011 IEEE International Conference on, 2011

Grasp Execution

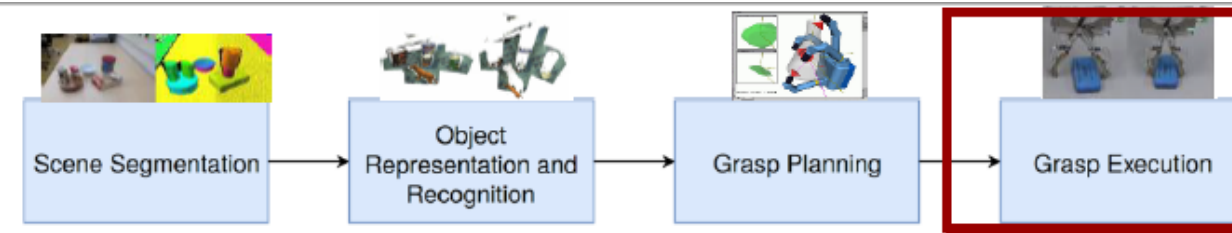


We have a segmented scene, a completed object, and a planned grasp, but how do we execute it?...

- Open Loop Grasp Execution



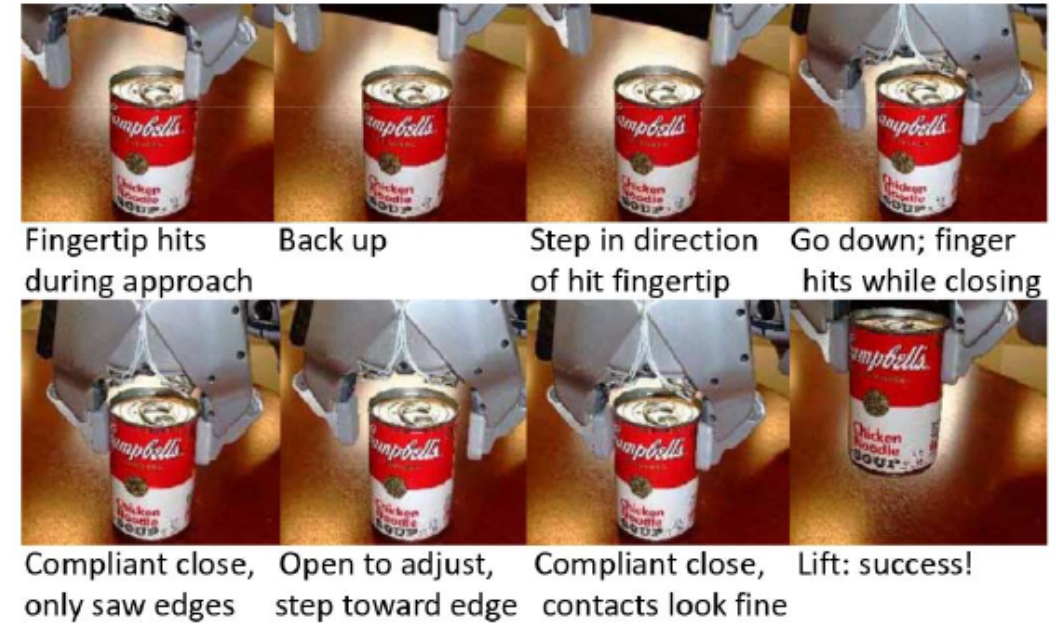
Grasp Execution



- Open Loop Grasp Execution is still mainstream
- No out of the box working solutions using feedback in general use
- Closest:

Hsiao, K., Chitta, S., Ciocarlie, M., & Jones, E. G.. Contact-reactive grasping of objects with partial shape information. In *Intelligent Robots and Systems (IROS), 2010 IEEE/RSJ International Conference*

Dang, Hao, and Peter K. Allen. "Stable grasping under pose uncertainty using tactile feedback." *Autonomous Robots* 36.4 (2014)



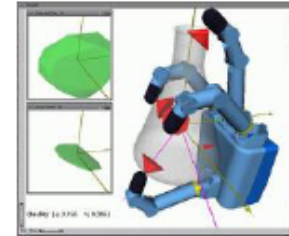
Pros:

- Able to integrate feedback

Cons:

- React poorly if object is perturbed in the process
- No vision
- heuristic

A Prototypical Grasping Pipeline: Now



Scene Segmentation

- Euclidean Cluster Extraction
- Object Discovery

Object
Representation and
Recognition

- RANSAC Instance matching
- Symmetry based completion

Grasp Planning

- Grasp Database
- Anneal through C_{space} via grasp quality heuristic
- Grasping Rectangles

Grasp Execution

- Open Loop Grasp Execution

General Problems:

Heuristics, Hand Crafted Features, Overly constrained, Small datasets, Little Sensory Feedback



How To Move Forward

- Problems: Heuristics, Hand Crafted Features, Overly Constrained, Small Datasets, Little Sensory Feedback
- Massive improvements in tangential fields in last 3 years:
 - **Big Data:** Significantly more available training data
 - **Simulation:** RGBD Rendering, Maintained contact during physics simulations
 - **Deep Learning:** Powerful classifiers
- Many of these improvements are being leveraged to alleviate current problems in grasping.



Big Data

Many of the approaches shown are heuristics validated on very small datasets.

- Are heuristics that work for these small dataset really representative?
- Difficult to develop data driven approaches if the data doesn't exist

Big Data

NYU Depth V2

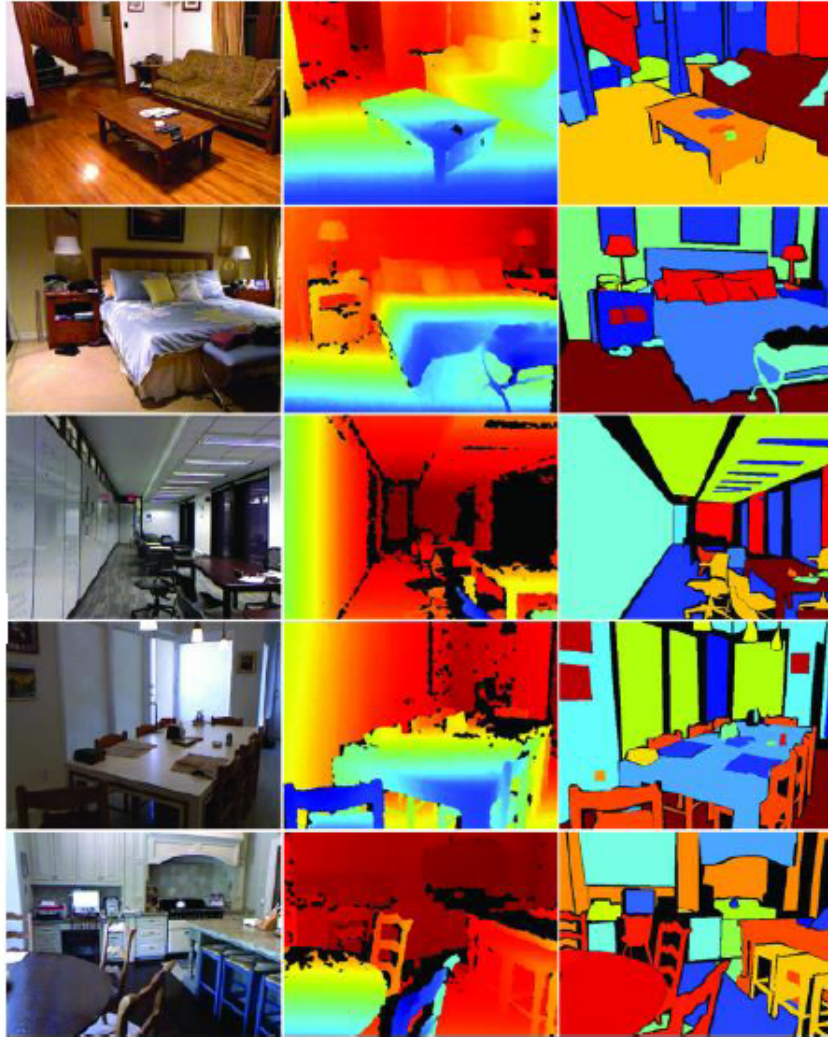
RGBD from Kinect

1449 densely labeled pairs
of aligned RGB and
depth images

407,024 new unlabeled frames

Each object is labeled with a class and an instance number (cup1, cup2, cup3, etc)

40 Categories



ShapeNet

Displaying 1 to 160 of 498



- 3 Million Models
- 220,000 categorized into 3135 categories (WordNet synsets)



Simulation

Part of the reason large datasets are slow to come into existence is because it requires a large amount of effort:

- Sensors change
- Takes time
- Often difficult to label ground truth

Simulation

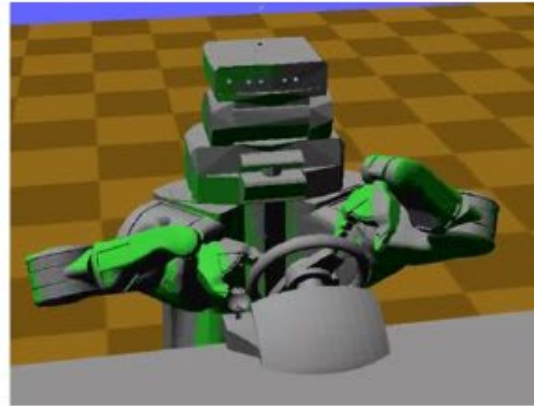
1) Embree: photo realistic rendering

Wald, Ingo, et al. "Embree: a kernel framework for efficient CPU ray tracing." *ACM Transactions on Graphics* 33.4 (2014).



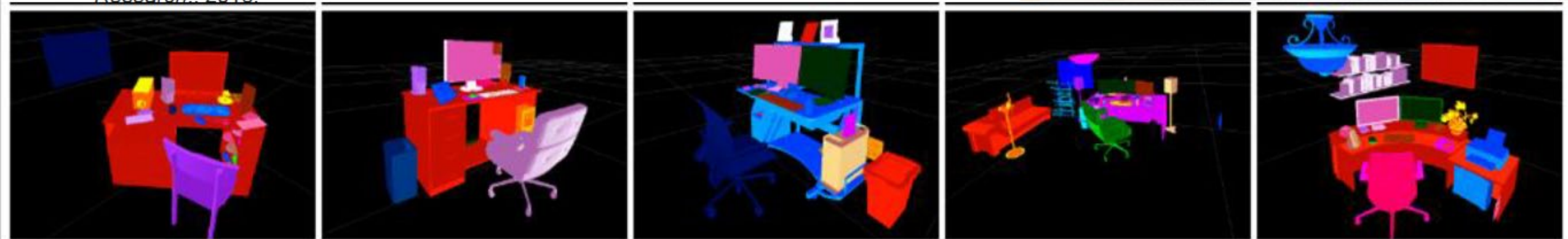
2) SceneNet: scene generation

Handa, Ankur, et al. "Scenetnet: Understanding real world indoor scenes with synthetic data." *arXiv preprint* (2015).



3) Klampt: contact simulation

Hauser, Kris. "Robust contact generation for robot simulation with unstructured meshes." *Robotics Research*, 2016.





Deep Learning

How to do data driven robotics:

- Before:
 - hand crafted features
 - Small datasets that work well with those features
- Now:
 - Let the network learn features from lots of data
 - Generate lots of data
 - Determine a good representation of the data

Deep Learning

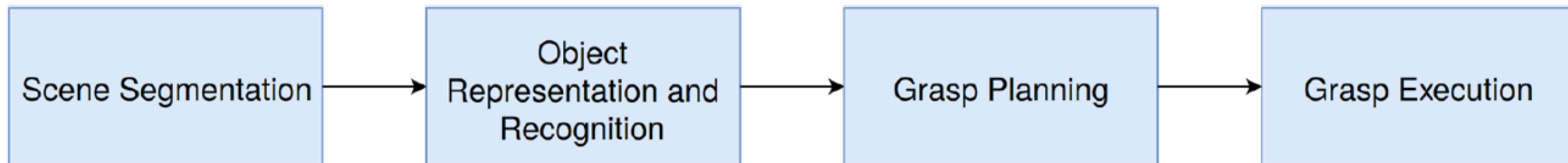
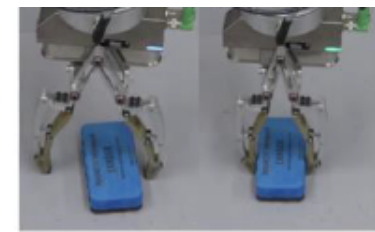
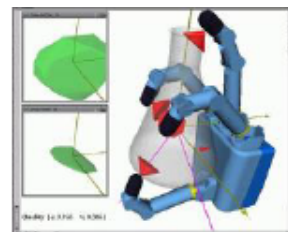
- ImageNet Challenge started in 2010:
 - 2012 winning team used deep learning
 - No Image Classification Task since 2014. Too easy
 - 2016:
 - [Object localization](#) for 1000 categories.
 - [Object detection](#) for 200 fully labeled categories.
 - [Object detection from video](#) for 30 fully labeled categories.
 - [Scene classification](#) for 365 scene categories
 - [Scene parsing](#)^{New} for 150 stu
- Nvidia Tesla K80 24GB g
 - 3D Convolutions



IN CS, IT CAN BE HARD TO EXPLAIN THE DIFFERENCE BETWEEN THE EASY AND THE VIRTUALLY IMPOSSIBLE.

<http://xkcd.com/1425/>

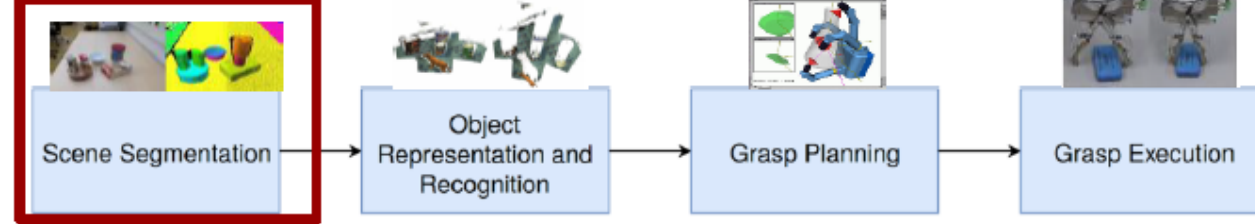
From 9/24/2014



A Prototypical Grasping Pipeline: 5 Years from now

- Dense RGBD Per Pixel Semantic Labeling
- Data driven scene and shape completion
- Learned grasp quality derived from simulation
- Learned closed loop torque control using visual and tactile feedback

Scene Segmentation



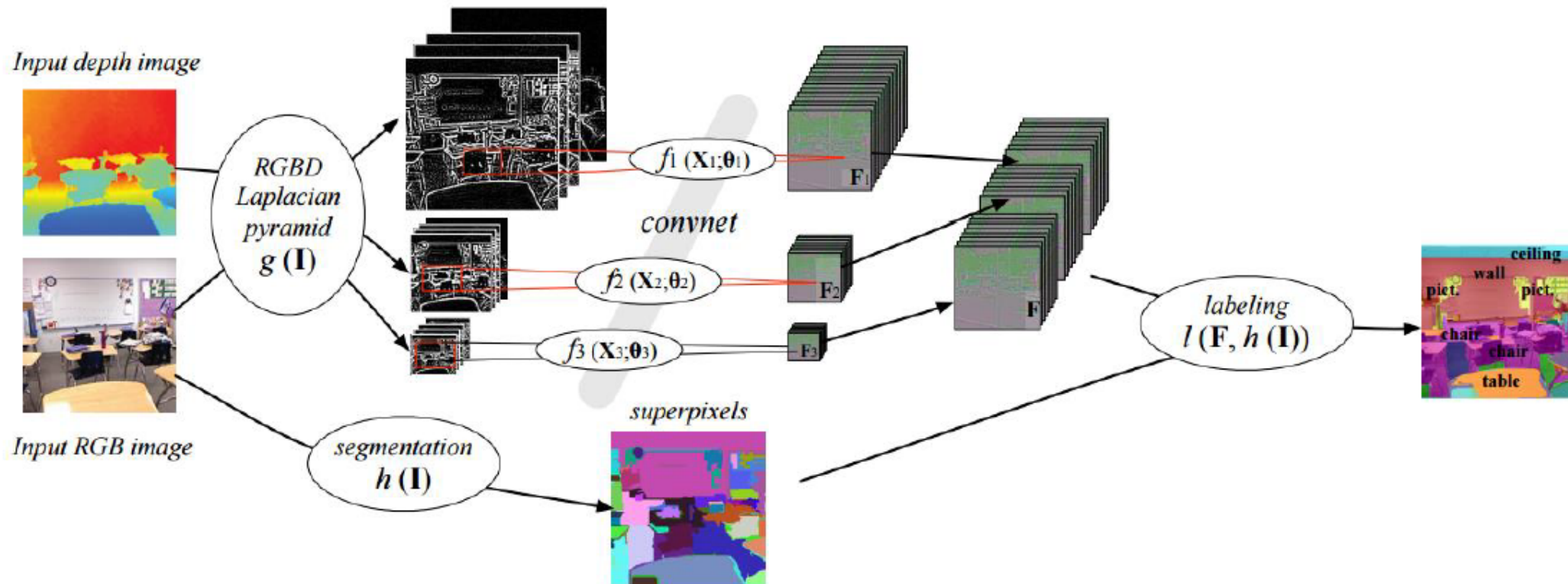
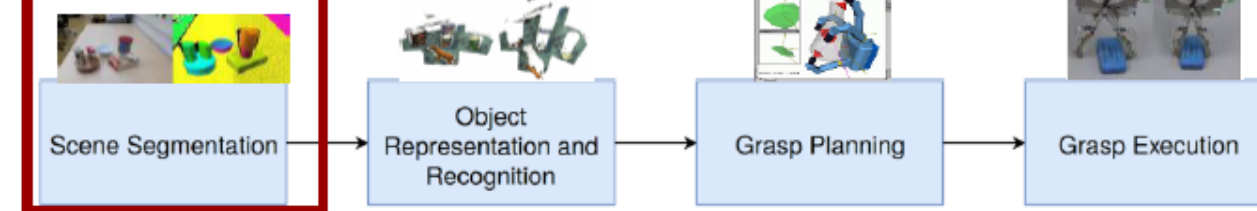
Before:

- Objectness detector
- PCL Euclidean cluster extraction

Now:

- Semantic per pixel/voxel/surflet labeling
- Powered by algorithms developed for ImageNet adapted to NYU-Depth V2 dataset.

Indoor Semantic Segmentation



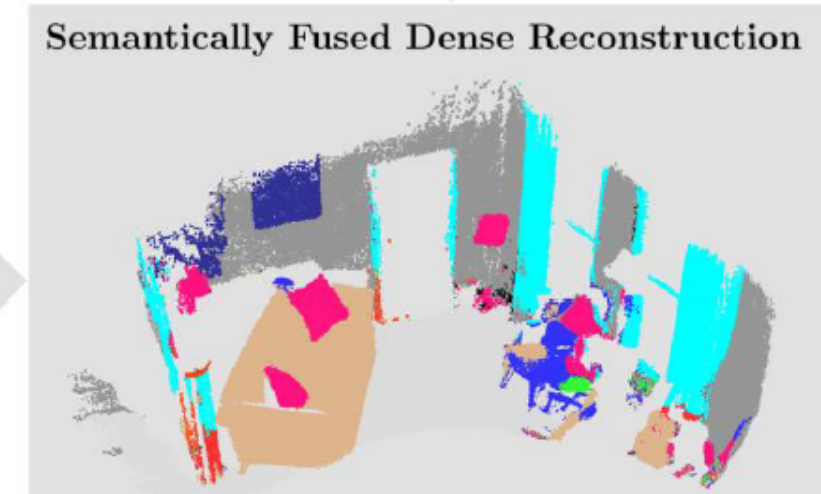
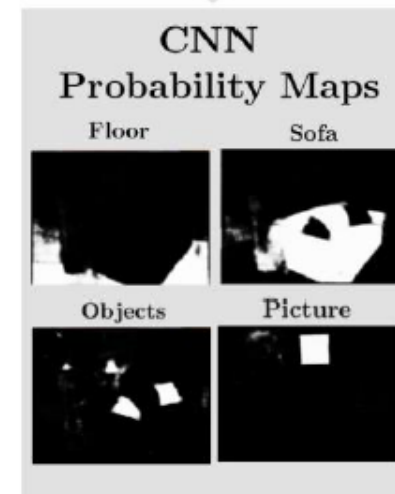
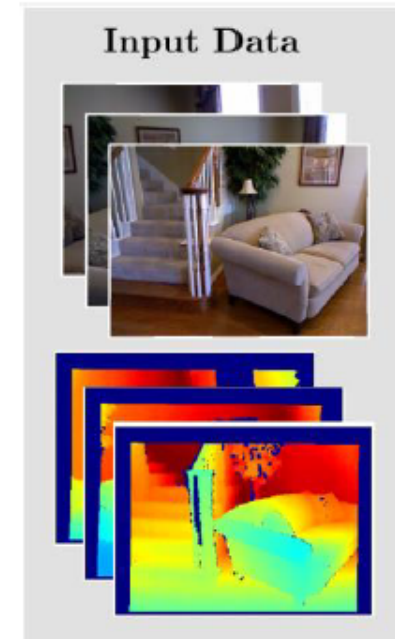
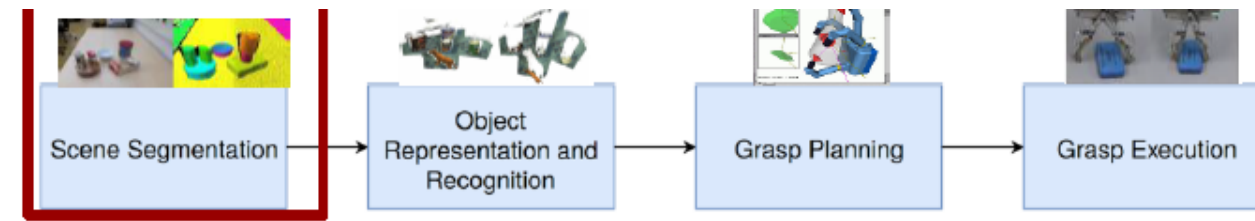
Couprie et al (NYU 2013): 52.4% per pixel accuracy with 16 categories

Long et al (UC Berkeley 2015): 65% per pixel accuracy with 40 categories

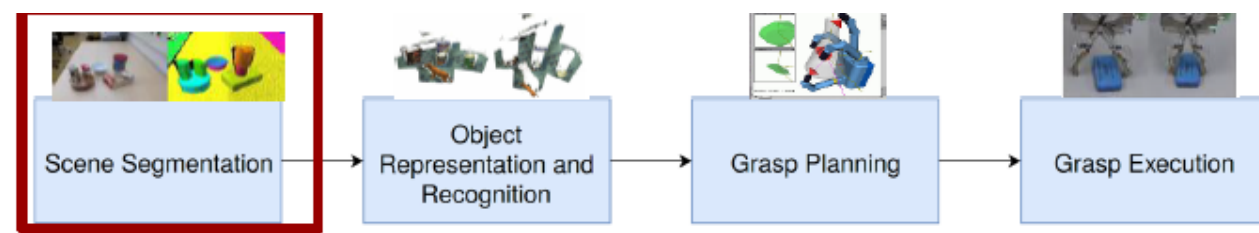
- Camille Couprie, Clement Farabet, Laurent Najman, and Yann LeCun. Indoor semantic segmentation using depth information. arXiv preprint arXiv:1301.3572, 2013
- Jonathan Long, Evan Shelhamer, and Trevor Darrell. Fully convolutional networks for semantic segmentation. In Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2015

Semantic Fusion

- Elastic Fusion Slam
- RGBD-CNN for per pixel labels
- Project pixels to surfels
- Bayesian update for per-surfel semantic label estimate



Scene Segmentation

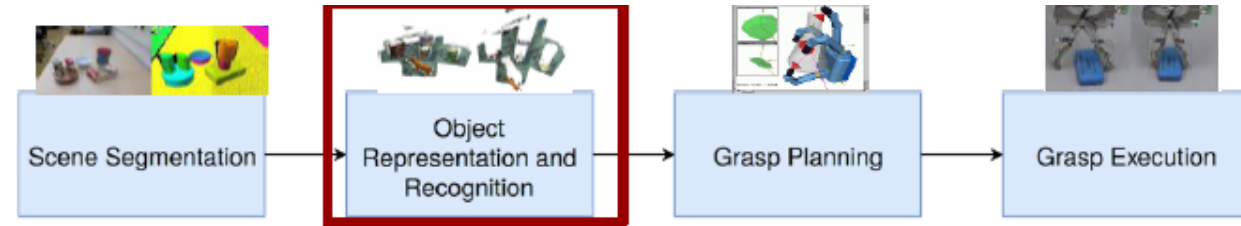


Deep Learning enabled per pixel labeling is here:

- Pros:
 - Dense labels
 - Semantic labels
 - Incredibly fast
- Current Hurdles:
 - We need more data! (Especially rgb-d data)
 - Object category is not interesting enough for robotics
 - Sensors improve and old datasets lose utility.



Object Modeling



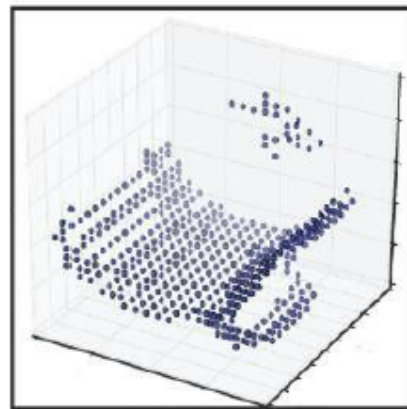
Before:

- Exact model matching approaches
- Simple symmetry and extrusion approaches for general completion

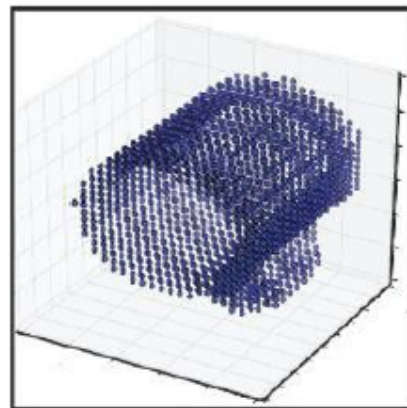
Now:

- Data Driven techniques for general completion

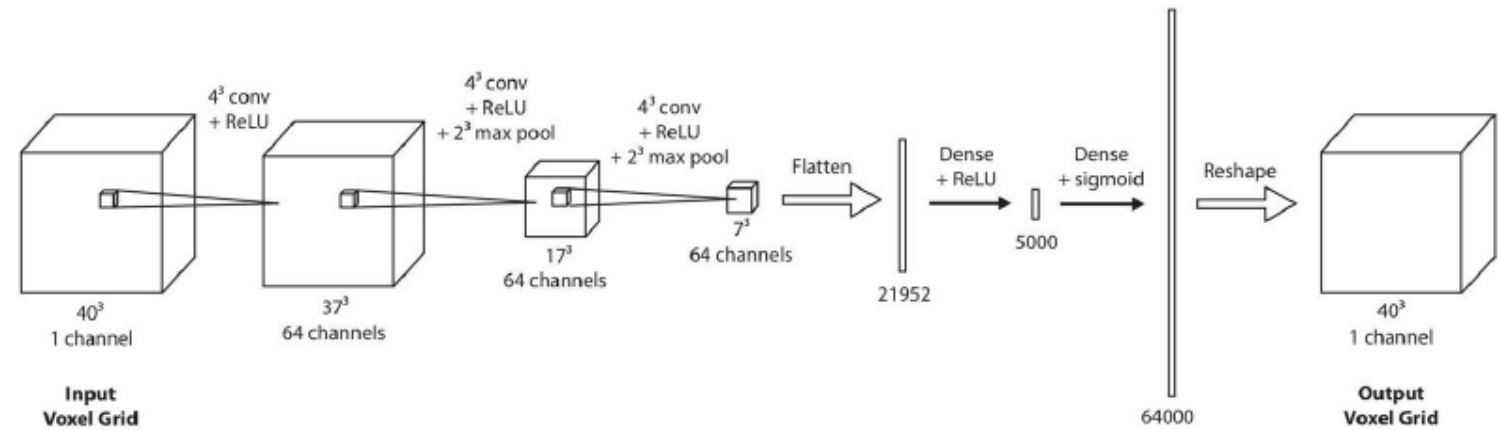
Shape Completion Enabled Robotic Grasping



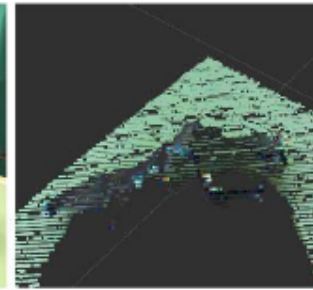
X



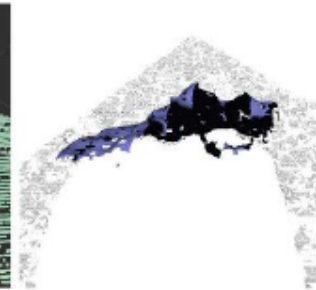
Y



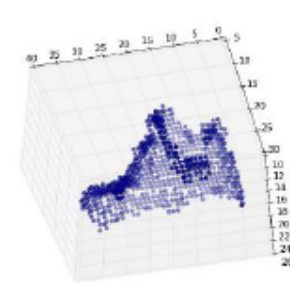
(a) Image of Occluded Side



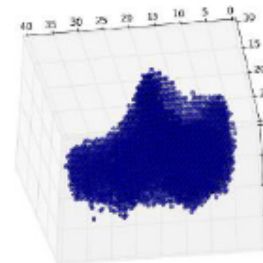
(b) Point Cloud



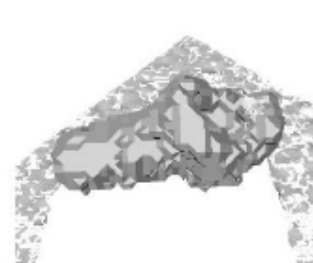
(c) Segmented and Meshed



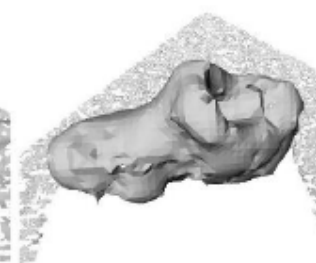
(d) CNN Input



(e) CNN Output



(f) Mesh



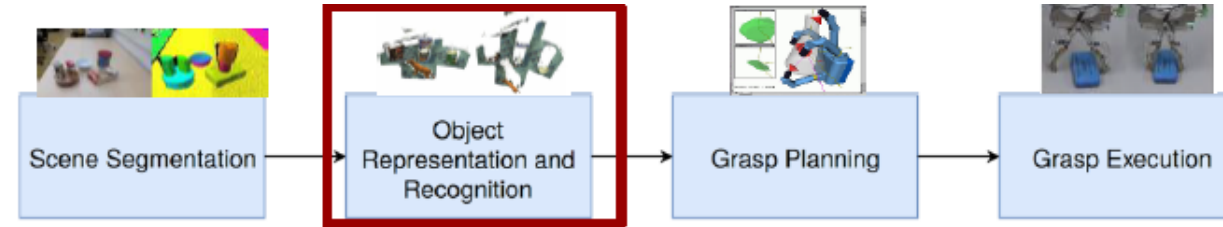
(g) Smoothed Mesh



(h) Grasp Planning

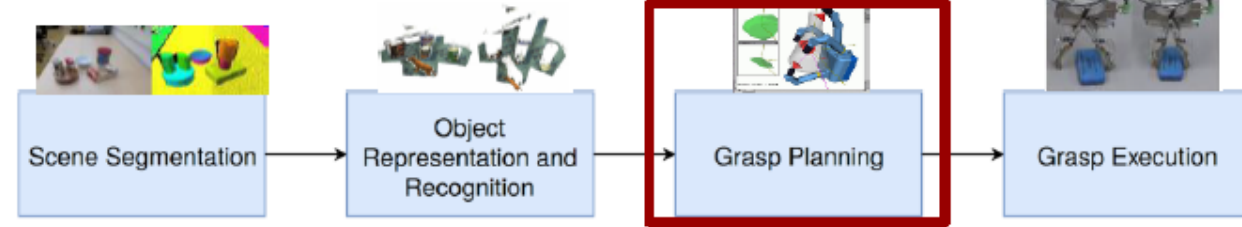


Object Modeling



- Yet to be a large scale dataset for general scene completion similar to NYU-Depth V2 dataset for semantic segmentation
- The 3d models exist:
 - ShapeNet: 3,000,000 models, 220,000 models out of which are classified into 3,135 categories

Grasp Planning



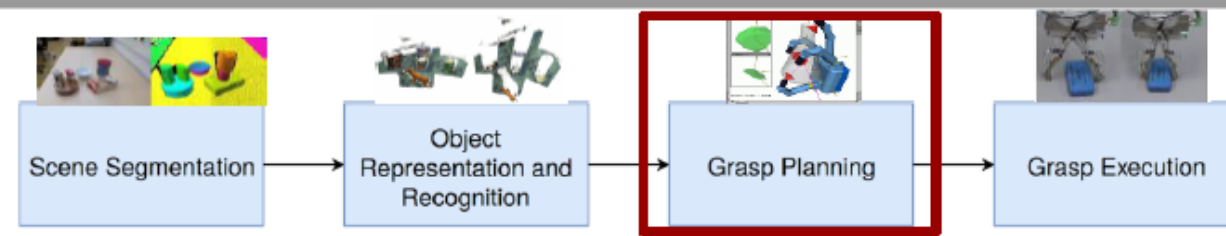
Before:

- Search in low dimensional space via handcrafted quality functions
- Database of objects and corresponding grasps
- Data driven parallel jaw grasps

Now:

- Deep Learning for data driven quality functions
- Simulated grasp executions to label training grasps

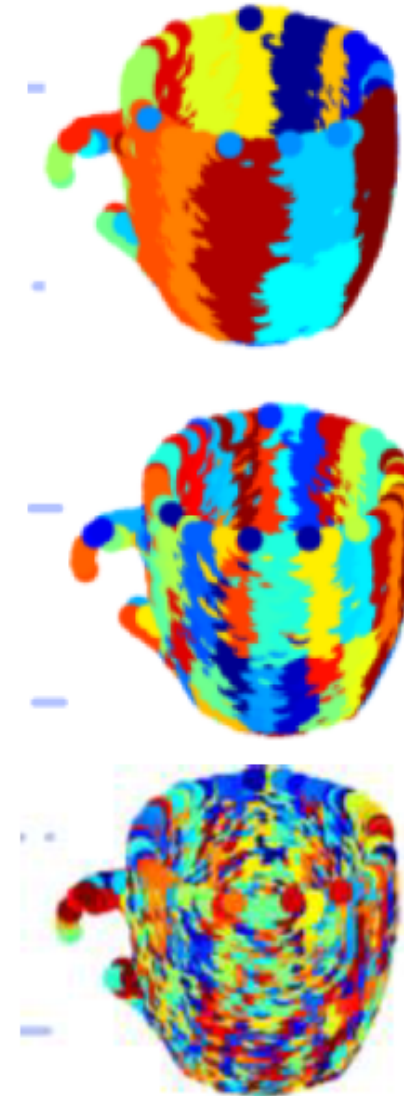
Hierarchical Fingertip space for multi-fingered precision grasping



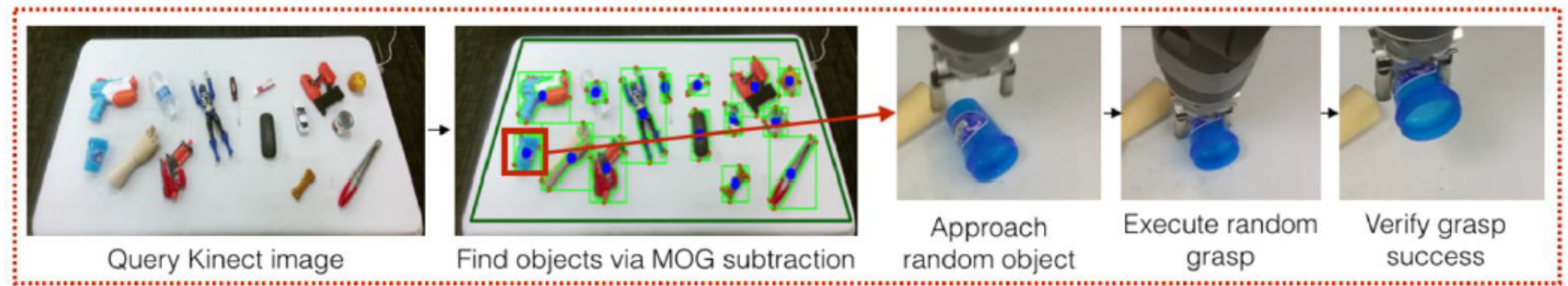
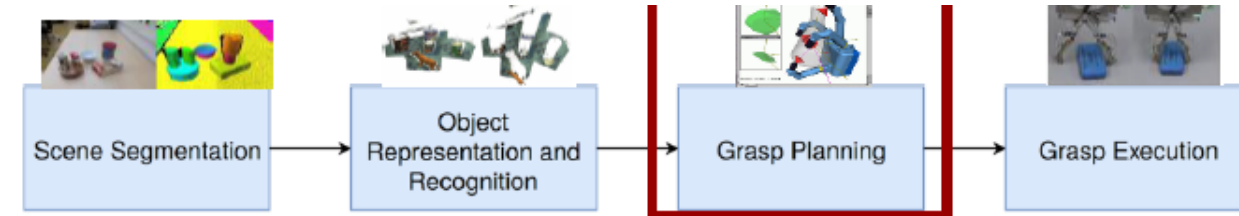
(A) extract a hierarchical fingertip space

(B) Using the fingertip space hierarchy and reachability, search for contacts and initial hand configuration.

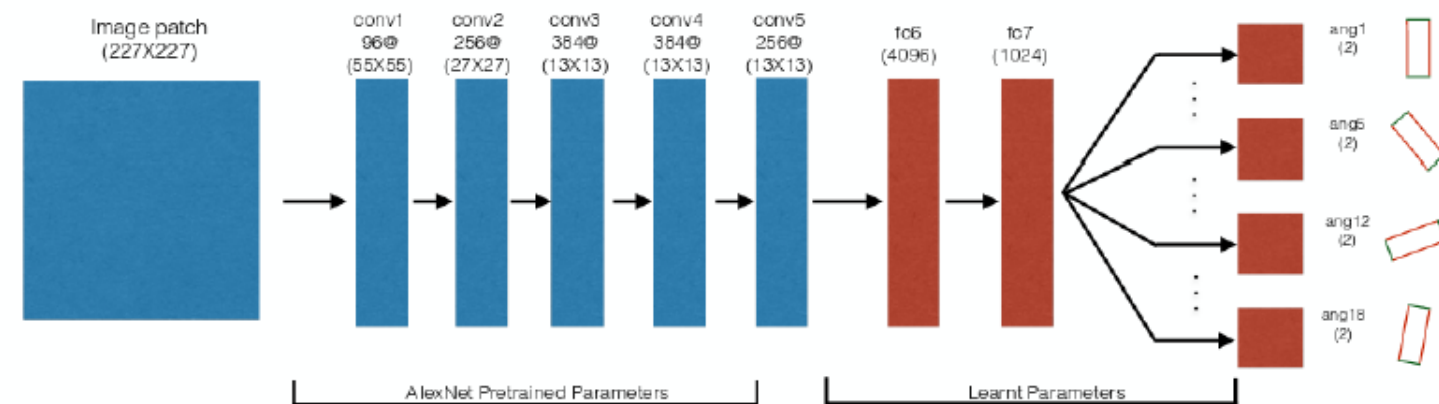
(C) grasp realized by local contact positions optimization with respect to the synthesized contacts



Supersizing self-supervision: Learning to grasp from 50K tries and 700 robot hours.



- RGB CNN
- 18 way binary classification
- 73% accuracy

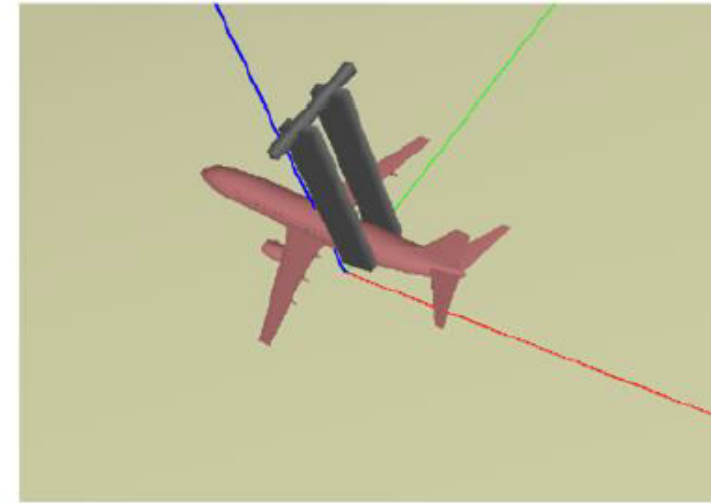
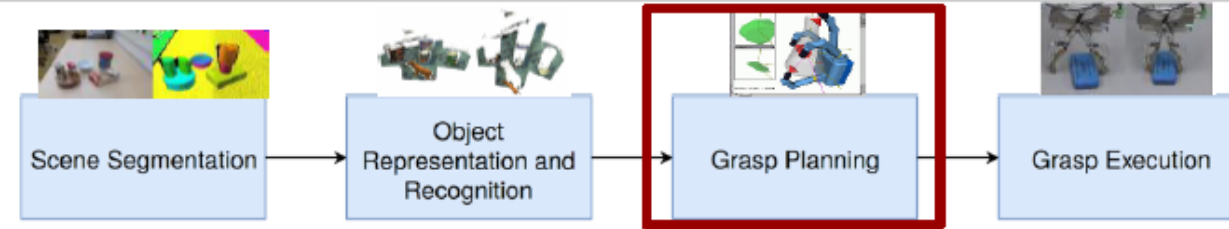


Lerrel Pinto and Abhinav Gupta. Supersizing self-supervision: Learning to grasp from 50k tries and 700 robot hours. arXiv preprint arXiv:1509.06825, 2015

Fig. 5. Our CNN architecture is similar to AlexNet [6]. We initialize our convolutional layers from ImageNet-trained Alexnet.

Deep Learning a grasp function for grasping under gripper pose uncertainty

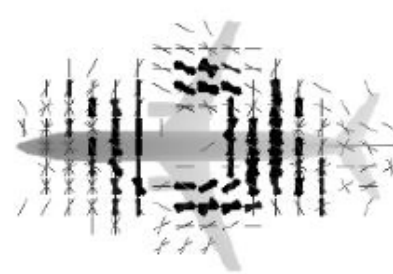
- Parallel Jaw Gripper grasps
- Supervised learning approach
- Training data evaluated with physics in simulation with gravity
- 80.3% accuracy



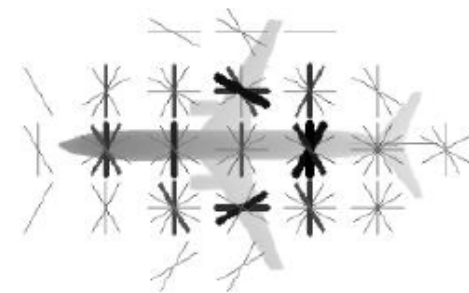
(a) Simulated depth image



(b) Simulated depth image after adding the noise model

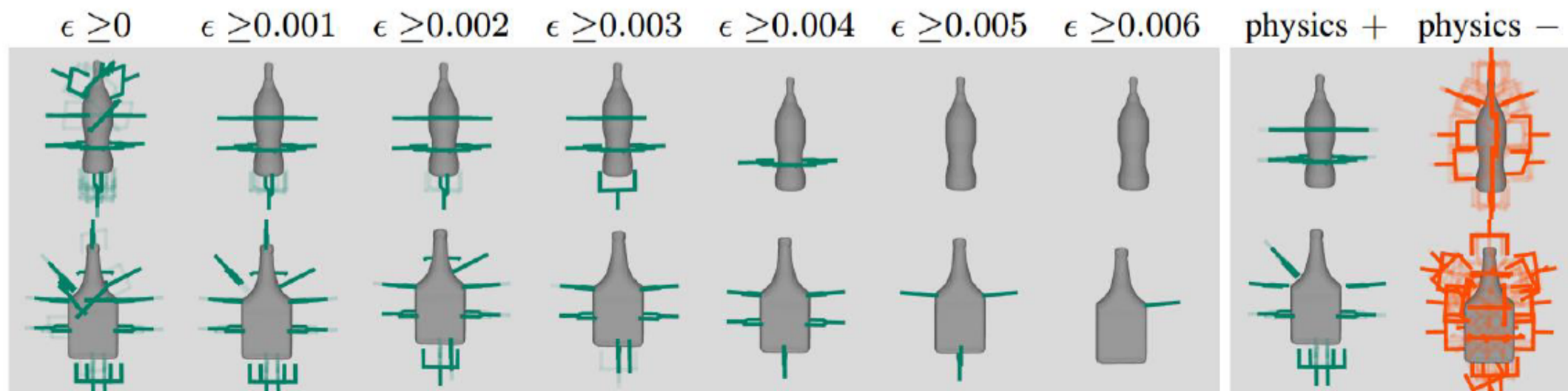
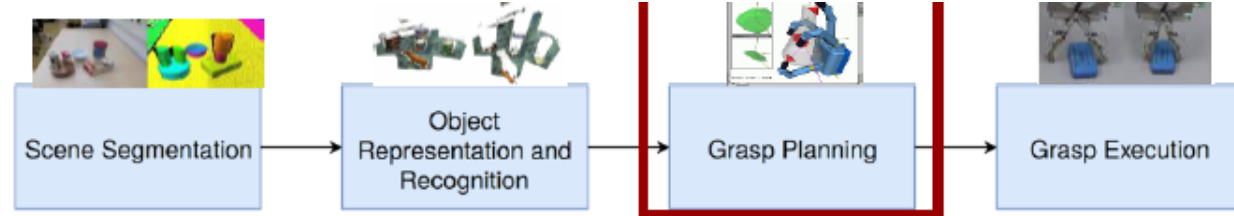


(c) Visualisation of the grasp function



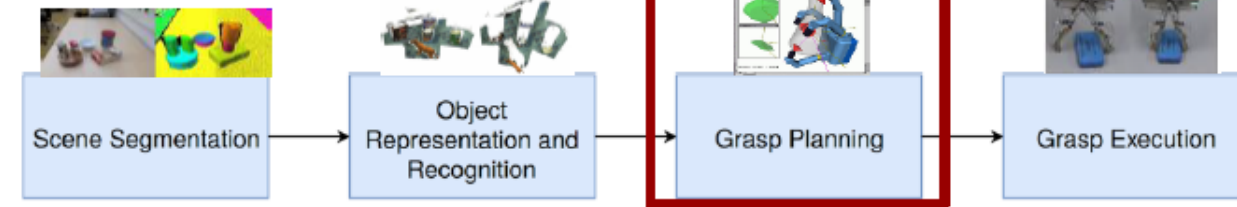
(d) Visualisation of the grasp function at a coarser scale

Leveraging big data for grasp planning



- Large scale database of parallel jaw gripper grasps
- Crowdsourcing show physics simulation better predictor to grasp success than epsilon-metric
- Train CNN to recognize good grasp locations

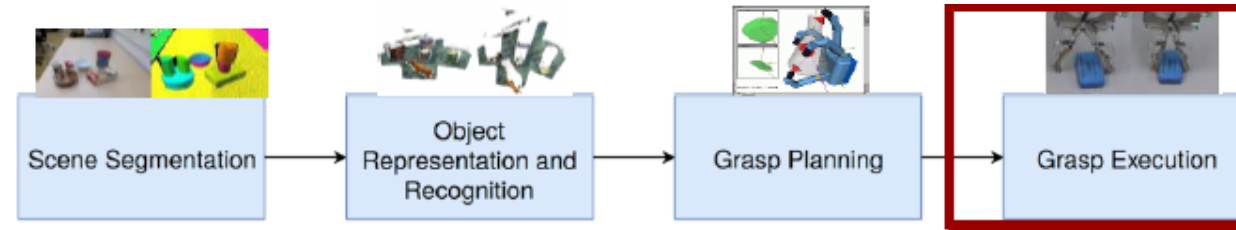
Grasp Planning



- Pros
 - Parallel Jaw Gripper from above is done
- Current Hurdles:
 - Higher dimensional grasp planning problems
 - Can we extend grasping rectangles?
 - Do we just need more efficient search algorithms?
- Possible Solution:



Grasp Execution



Before:

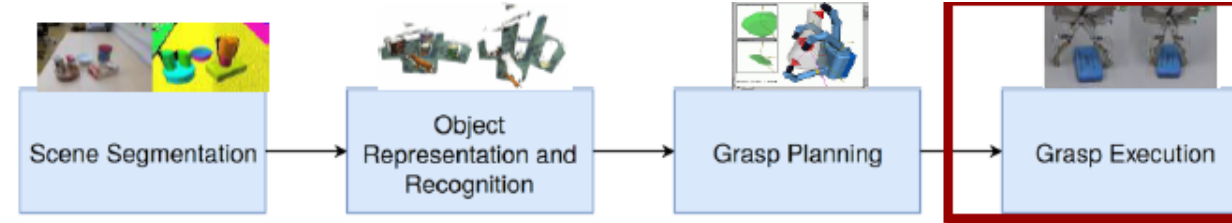
- Open Loop Grasp Execution

Now:

- Deep learning enabled mapping from sensory information to movement

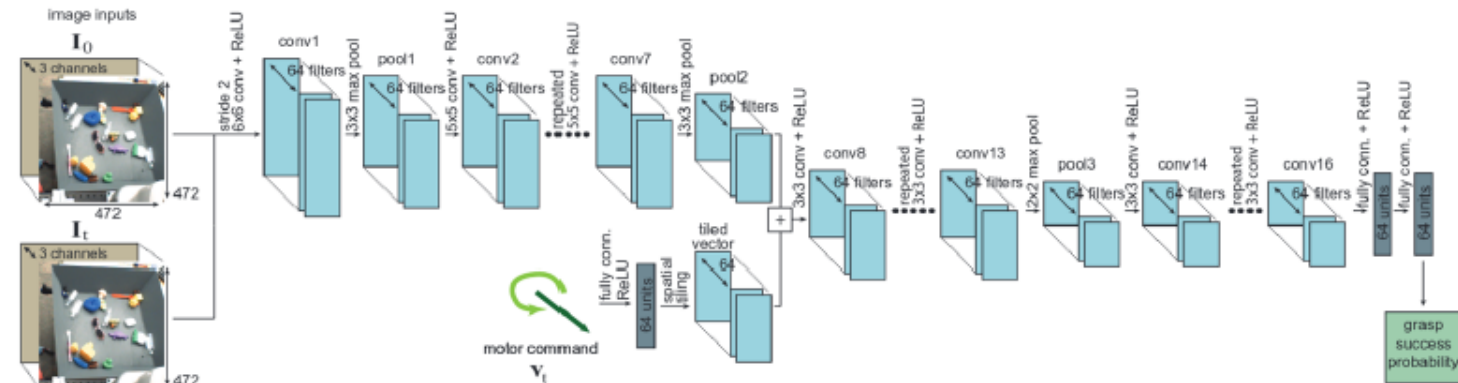
Learning Hand-Eye Coordination for Robotic Grasping with Deep Learning and Large-Scale Data Collection

- CNN to predict the probability that task-space motion of the gripper will result in a successful grasp
- Servoing algorithm powered by the CNN



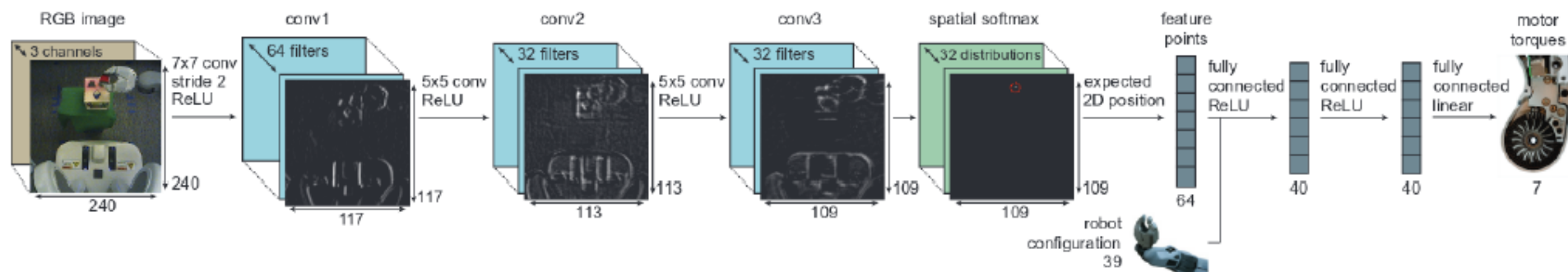
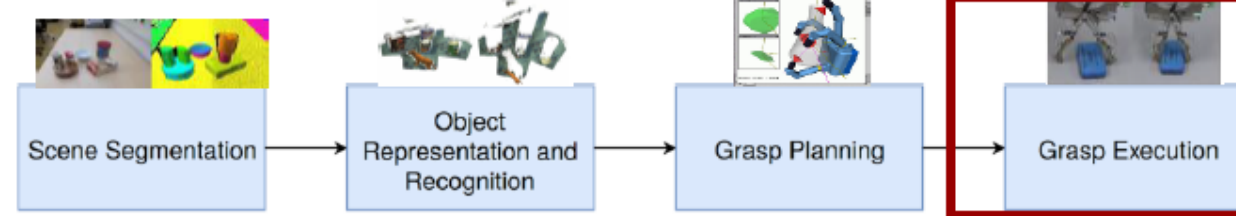
Algorithm 1 Servoing mechanism $f(\mathbf{I}_t)$

- 1: Given current image $\mathbf{I}_t$ and network g .
- 2: Infer $\mathbf{v}_t^*$ using g and CEM.
- 3: Evaluate $p = g(\mathbf{I}_t, \emptyset) / g(\mathbf{I}_t, \mathbf{v}_t^*)$.
- 4: **if** $p > 0.9$ **then**
- 5: Output $\emptyset$, close gripper.
- 6: **else if** $p \leq 0.5$ **then**
- 7: Modify $\mathbf{v}_t^*$ to raise gripper height and execute $\mathbf{v}_t^*$.
- 8: **else**
- 9: Execute $\mathbf{v}_t^*$.
- 10: **end if**

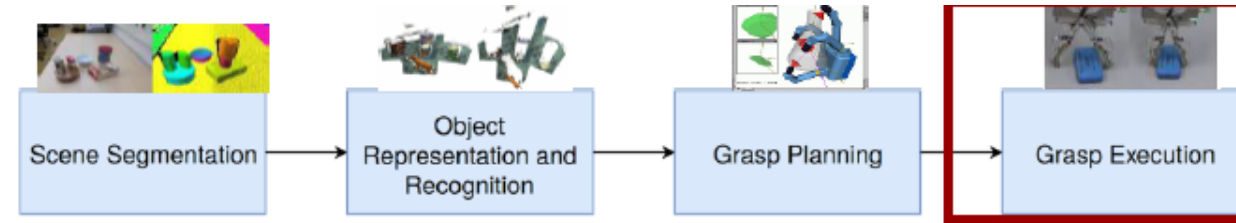


End-to-end training of deep visuomotor policies

- CNN mapping raw images -> torques
- Train several Linear Gaussian Controllers choose an action given full scene info (exact object pose, end effector pose). Different LGC for different start configurations
- Training a CNN to replicate the linear gaussian controllers. Using observations (image, + encoder values)



Towards Adapting Deep Visuomotor Representations from Simulated to Real Environments



An initial step toward pretraining deep visuomotor policies entirely in simulation, significantly reducing physical demands when learning complex policies

3 Loss terms:

- 1) Standard pose estimation loss
- 2) Domain confusion loss to align the synthetic and real domains in feature space.
- 3) contrastive loss to align specific pairs in feature space

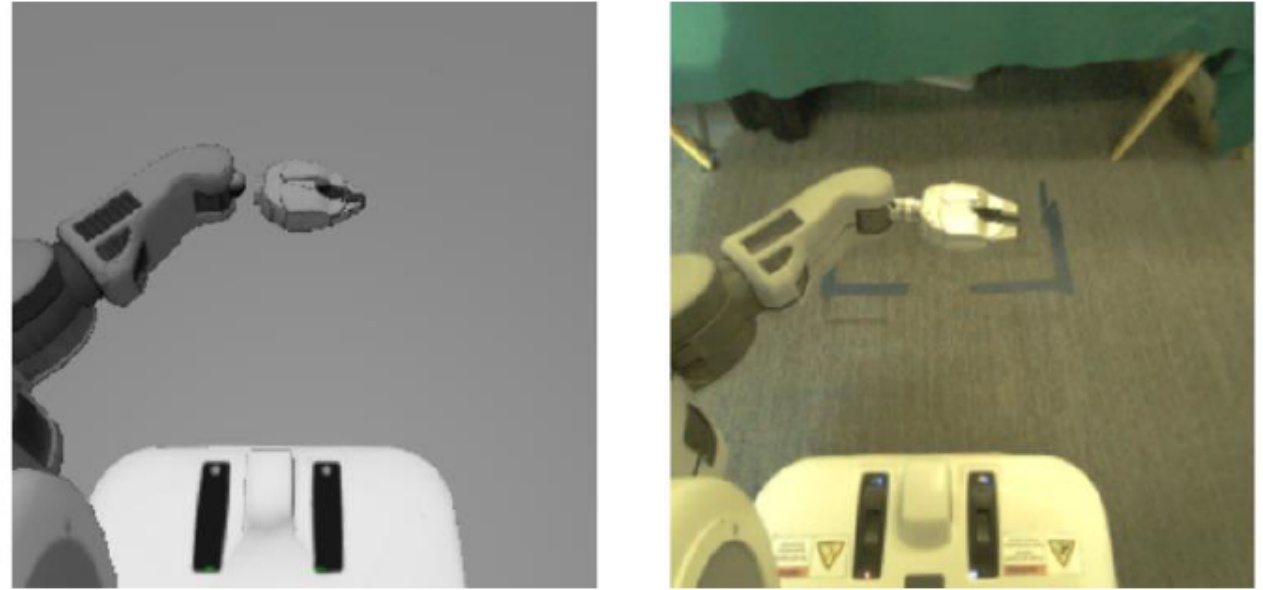
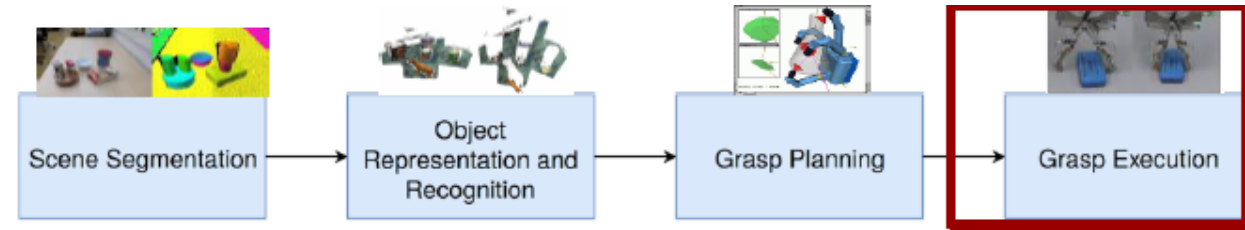


Fig. 3. A pair of corresponding synthetic (left) and real-world (right) images used for our pose estimation evaluation. The task is to predict the 3D pose of the PR2's gripper.



Grasp Execution



- Pros
 - Simple Controllers possible to train
- Current Hurdles:
 - Each controller learns to reach a very specific goal
- Possible Solutions
 - Have lots of controllers for different canonical grasp types for different objects.
 - Train them in simulation

Future Research Directions

- Segmentation:
 - No per pixel labeled rgb-d dataset geared towards robotics
- Object Modeling:
 - No large scale dataset with partial observations and ground truth model with pose
- Grasp Planning:
 - Multi-fingered grasp planning integrating:
 - Anneal over Hierarchical Fingertip or EigenGrasp space



How to grasp an object / objects with a robotic hand?

Analytical Models



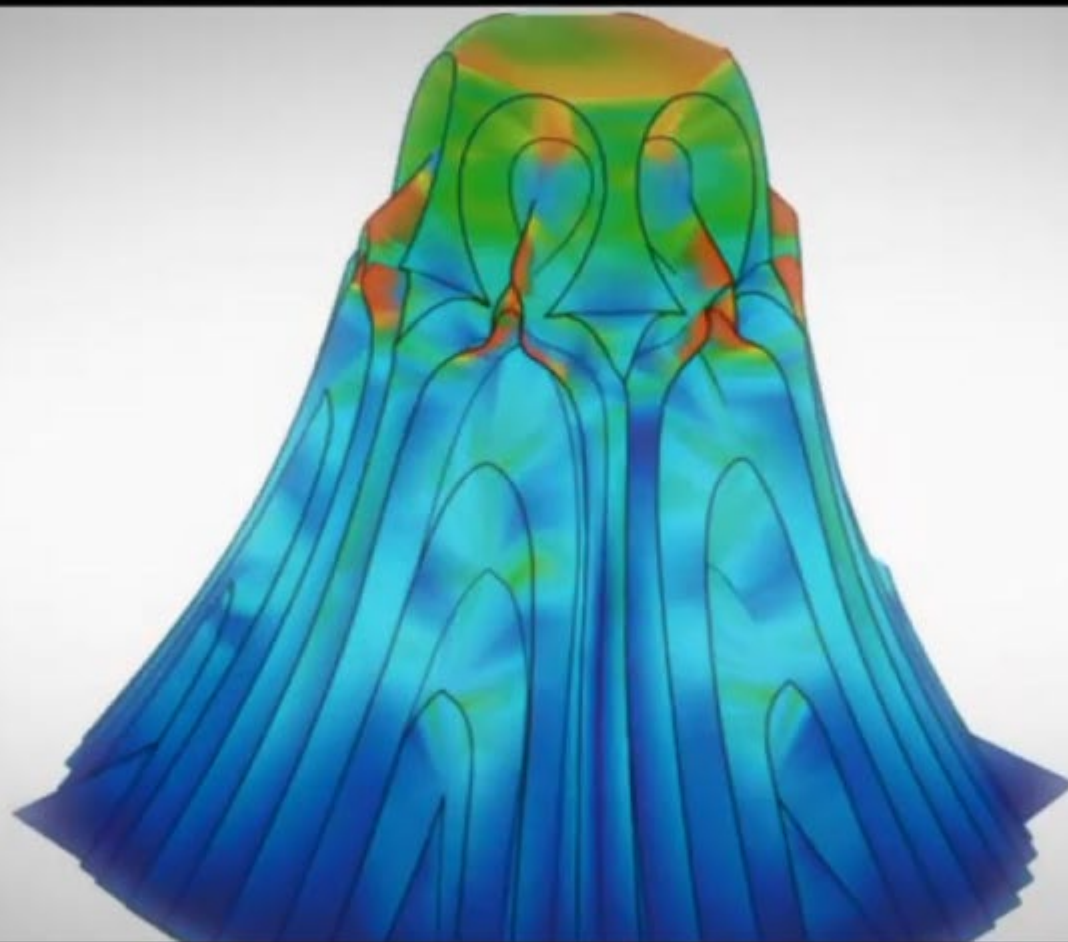
2000-2016

Large Model





Any new hand ?



施加振荡磁场可以实现往复运动

秒剪 ▶ @冲





Cite this: *Soft Matter*, 2023, 19, 2186

Shape memory meta-laminar jamming actuators fabricated by 4D printing

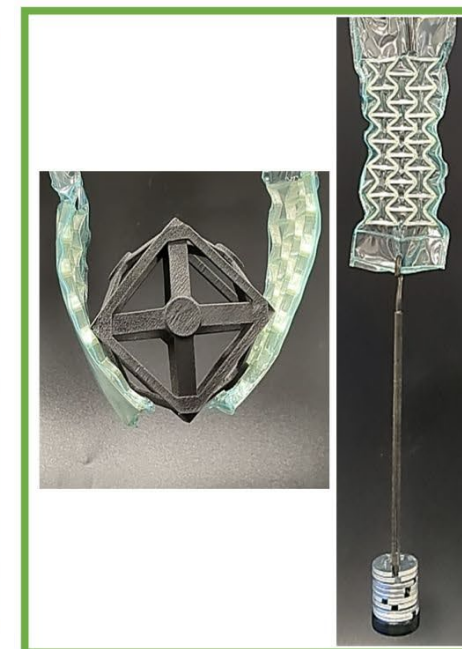
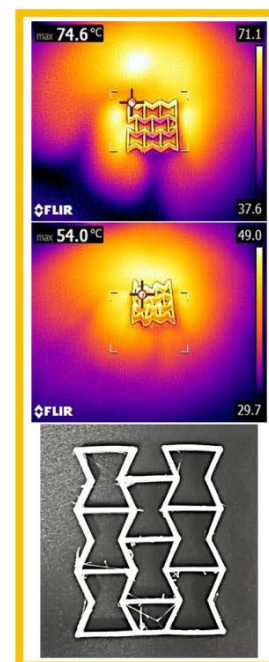
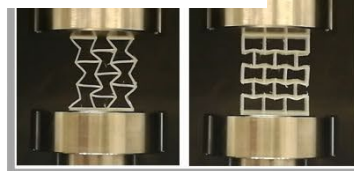
Mohammadreza Lalegani Dezaki  and Mahdi Bodaghi *

Laminar jamming (LJ) technology is a hot topic because it allows for the transition from conventionally quick, precise, and high-force rigid robots to flexible, agile, and secure soft robots. This article introduces a novel conceptual design of meta-laminar jamming (MLJ) actuators with a polyurethane shape memory polymer (SMP)-based meta-structure fabricated by 4D printing (4DP). The sustainable MLJ actuators behave as soft/hard robots via hot and cold programming accompanied by negative air pressure. The advantage of MLJ actuators over conventional LJ actuators is that a continuous negative air pressure is not required to stimulate the actuator. SMP meta-structures with circle, rectangle, diamond, and auxetic shapes are 4D printed. Mechanical properties of the structures are evaluated through three-point bending and compression tests. Shape memory effects (SMEs) and shape recovery of meta-structures and MLJ actuators are investigated via hot air programming. MLJ actuators with auxetic meta-structure cores show a better performance in terms of contraction and bending with 100% shape recovery after stimulation. The sustainable MLJ actuators have the capabilities of shape recovery and shape locking with zero input power while holding 200 g weight. The actuator can easily lift and hold objects of varying weights and shapes without requiring any power input. This actuator has demonstrated its versatility in potential applications, such as functioning as an end-effector and a gripper device.

Received 25th January 2023,
Accepted 11th February 2023

DOI: 10.1039/d3sm00106g

rsc.li/soft-matter-journal



l Test

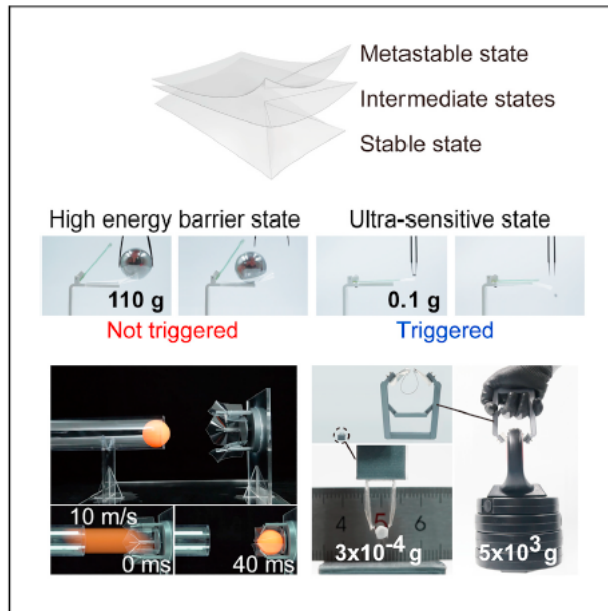
Shape Recovery

MLJ Actuators
Shape Locking

Applications

Article

Ultra-tunable bistable structures for universal robotic applications



Jiang et al. report a type of bistable structure that can be precisely tuned to ultra-sensitive states by bringing the structures on the verge of instability. Accordingly, the trigger force required to induce the fast snap-through process can be controlled to 0.1% of its maximum value.

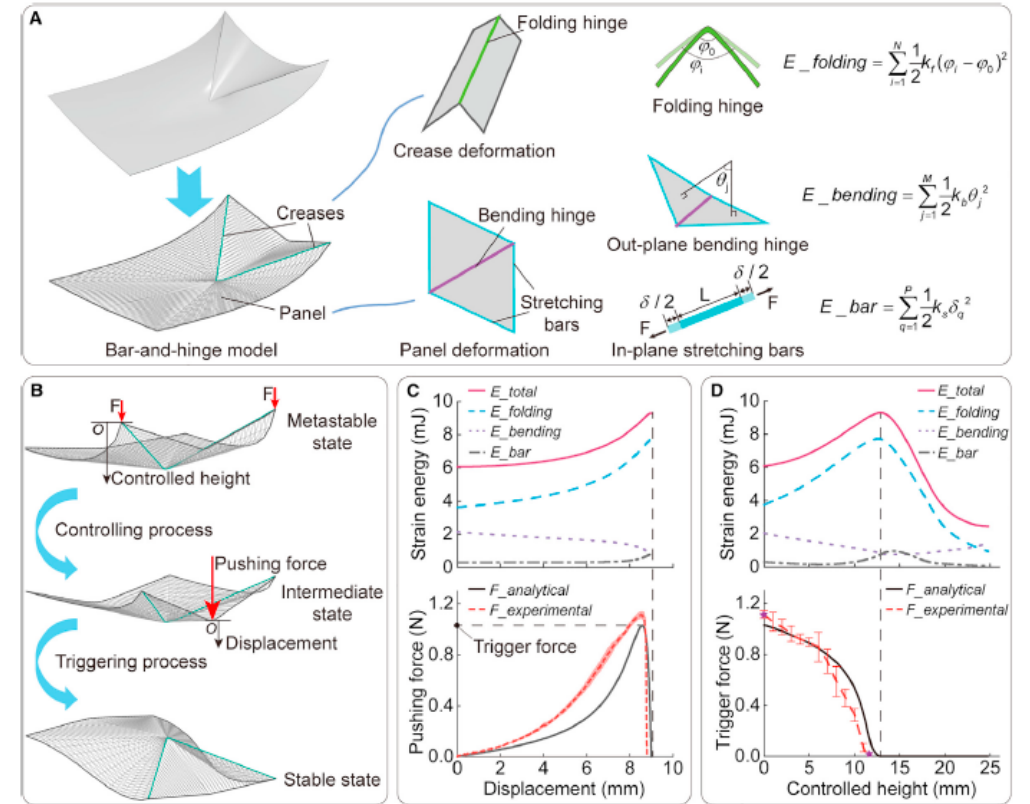
Yongkang Jiang, Yingtian Li, Ke Liu, ..., Diansheng Chen, Lei Wang, Jamie Paik
yt.li@sia.ac.cn

Highlights

A general type of ultra-tunable bistable structure is proposed

The structures are applicable for varied materials, scales, actuation, etc.

The structures are demonstrated in various robotic applications





- Levine et al. "Learning hand-eye coordination for robotic grasping with deep learning and large-scale data collection." *The International Journal of Robotics Research* 37.4-5 (2018): 421-436.
- Peng et al. "Sim-to-real transfer of robotic control with dynamics randomization." *arXiv preprint arXiv:1710.06537* (2017).
- Tobin et al. "Domain randomization for transferring deep neural networks from simulation to the real world." *Intelligent Robots and Systems (IROS), 2017 IEEE/RSJ International Conference on*. IEEE, 2017.
- Rajeswaran et al. "Learning complex dexterous manipulation with deep reinforcement learning and demonstrations." *arXiv preprint arXiv:1709.10087* (2017).
- Kumar, Vikash, Emanuel Todorov, and Sergey Levine. "Optimal control with learned local models: Application to dexterous manipulation." *Robotics and Automation (ICRA), 2016 IEEE International Conference on*. IEEE, 2016.
- Schulman et al. "Proximal policy optimization algorithms." *arXiv preprint arXiv:1707.06347* (2017).
- Sadeghi & Levine. "CAD2RL: Real single-image flight without a single real image." *arXiv preprint arXiv:1611.04201* (2016).

162
SHARES

Facebook



Twitter



Pinterest



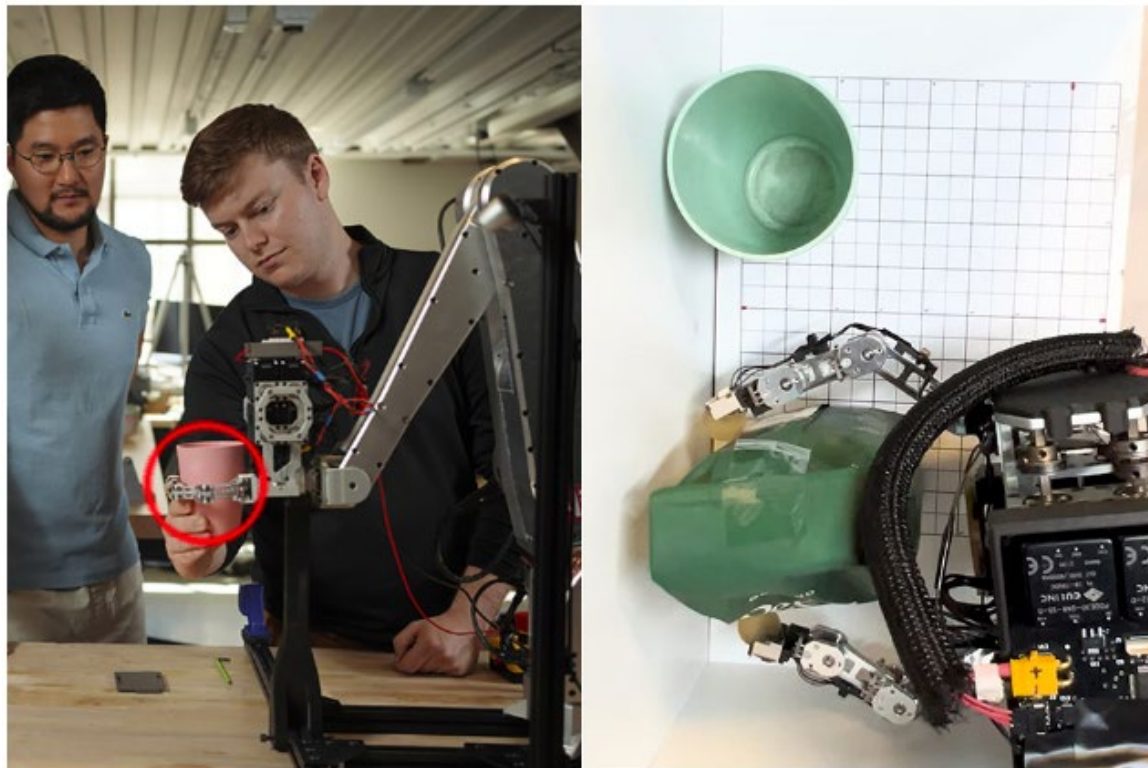
LinkedIn



Reddit



WhatsApp



MIT engineers have built a robotic gripper capable of grasping objects purely by reflex, adapting in real-time to roll, palm or pinch an object to get a more secure hold. The system consists of a high-speed arm, two lightweight, multi-jointed fingers, a camera mounted to the base of the arm, and custom high-bandwidth sensors at the fingertips.



SUBSCRIBE

Speedy robo-gripper reflexively organizes cluttered spaces

Rather than start from scratch after a failed attempt, the pick-and-place robot adapts in the moment to get a better hold.



Watch Video

Jennifer Chu | MIT News Office

April 27, 2023





Towards Robust Autonomous Grasping with Reflexes Using High-Bandwidth Sensing and Actuation

Andrew SaLoutos¹, Hongmin Kim¹, Elijah Stanger-Jones¹, Menglong Guo¹, and Sangbae Kim¹

Abstract—Modern robotic manipulation systems fall short of human manipulation skills partly because they rely on closing feedback loops exclusively around vision data, which reduces system bandwidth and speed. By developing autonomous grasping reflexes that rely on high-bandwidth force, contact, and proximity data, the overall system speed and robustness can be increased while reducing reliance on vision data. We are developing a new system built around a low-inertia, high-speed arm with nimble fingers that combines a high-level trajectory planner operating at less than 1 Hz with low-level autonomous reflex controllers running upwards of 300 Hz. We characterize the reflex system by comparing the volume of the set of successful grasps for a naive baseline controller and variations of our reflexive grasping controller, finding that our controller expands the set of successful grasps by 55% relative to the baseline. We also deploy our reflexive grasping controller with a simple vision-based planner in an autonomous clutter clearing task, achieving a grasp success rate above 90% while clearing over 100 items.

I. INTRODUCTION

Achieving human-like versatility in robotic manipulation will depend on developing hands that are as nimble and reactive as human hands. Much work has been done on

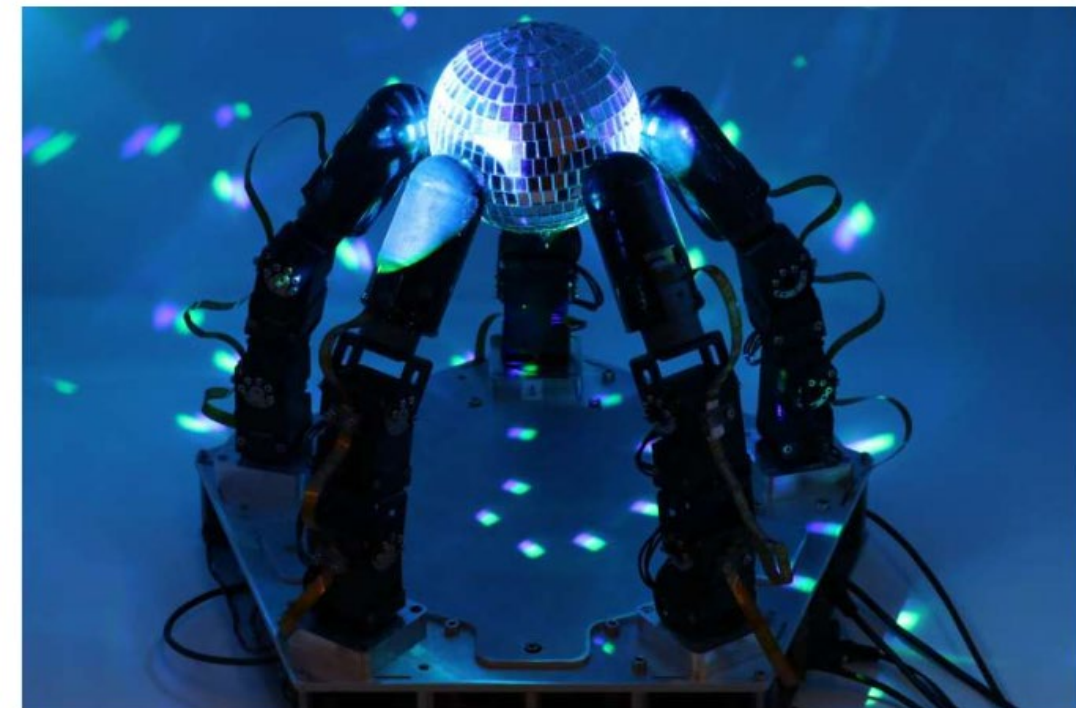


Fig. 1. Manipulation platform with reflexes for autonomous grasping. Our reflexive grasping controllers utilize our manipulation platform’s high-bandwidth actuation and low-latency sensing modalities to perform autonomous grasping tasks, such as clearing a cluttered shelf.

disturbing the object to be grasped can impose harsh constraints on the entire manipulation plan, resulting in slow or conservative actions. The dependence of many robotic manipulation systems on vision data exacerbates this challenge.

Highly dexterous robot hand can operate in the dark—just like us

by Columbia University School of Engineering and Applied Science



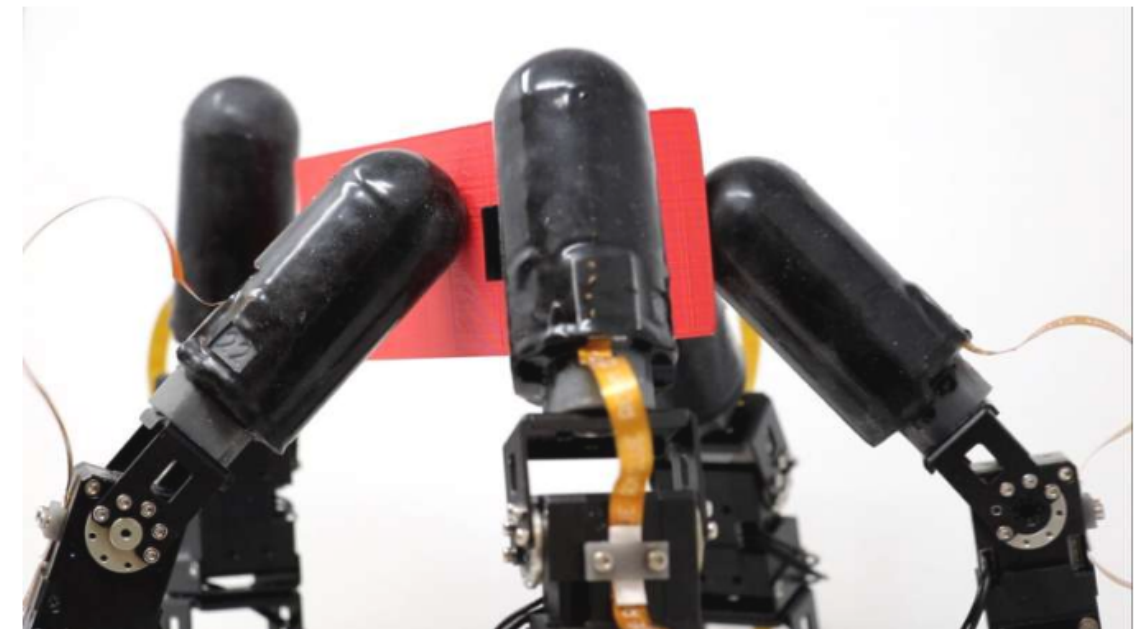
Using a sense of touch, a robot hand can manipulate in the dark, or in difficult lightin...

Think about what you do with your hands when you're home at night pushing buttons on your TV's remote control, or at a restaurant using all kinds of cutlery and glassware. These skills are all based on touch, while you're watching a TV program or choosing something from the menu. Our hands and fingers are incredibly skilled mechanisms, and highly sensitive to boot.

Teaching the hand to perform complex tasks

For this new work, led by Clocarlie's doctoral researcher, Gagan Khandate, the researchers designed and built a robot hand with five fingers and 15 independently actuated joints—each finger was equipped with the team's touch-sensing technology.

The next step was to test the ability of the tactile hand to perform complex manipulation tasks. To do this, they used new methods for motor learning, or the ability of a robot to learn new physical tasks via practice. In particular, they used a method called deep reinforcement learning, augmented with new algorithms that they developed for effective exploration of possible motor strategies.



Machine learning algorithms process the data from the tactile sensors to produce co...

Robot completed approximately one year of practice in only hours of real-time

A robotic arm with four fingers is shown holding a glowing, segmented sphere. The sphere is composed of many small, rectangular segments that are illuminated with a bright, multi-colored light. The arm is dark and mechanical, with visible joints and wiring. The background is dark with some colorful light spots.

DEXTEROUS MANIPULATION WITH TACTILE FINGERS

Gagan Khandate
Siqi Shang
Eric Chang
Tristan Saidi
Johnson Adams
Matei Ciocarlie



COLUMBIA ENGINEERING



Sampling-based Exploration for Reinforcement Learning of Dexterous Manipulation

Gagan Khandate^{*†}, Siqi Shang^{*†}, Eric T. Chang[‡], Tristan Luca Saidi[†], Johnson Adams[‡] and Matei Ciocarlie[‡]

[†]Dept. of Computer Science [‡]Dept. of Mechanical Engineering ^{*}joint first authorship

Columbia University, New York, NY 10027, USA

Corresponding email: gagank@cs.columbia.edu

Abstract—In this paper, we present a novel method for achieving dexterous manipulation of complex objects, while simultaneously securing the object without the use of passive support surfaces. We posit that a key difficulty for training such policies in a Reinforcement Learning framework is the difficulty of exploring the problem state space, as the accessible regions of this space form a complex structure along manifolds of a high-dimensional space. To address this challenge, we use two versions of the non-holonomic Rapidly-Exploring Random Trees algorithm; one version is more general, but requires explicit use of the environment’s transition function, while the second version uses manipulation-specific kinematic constraints to attain better sample efficiency. In both cases, we use states found via sampling-based exploration to generate reset distributions that enable training control policies under full dynamic constraints via model-free Reinforcement Learning. We show that these policies are effective at manipulation problems of higher difficulty than previously shown, and also transfer effectively to real robots. These videos can be found on the project website: sbri.cs.columbia.edu

I. INTRODUCTION

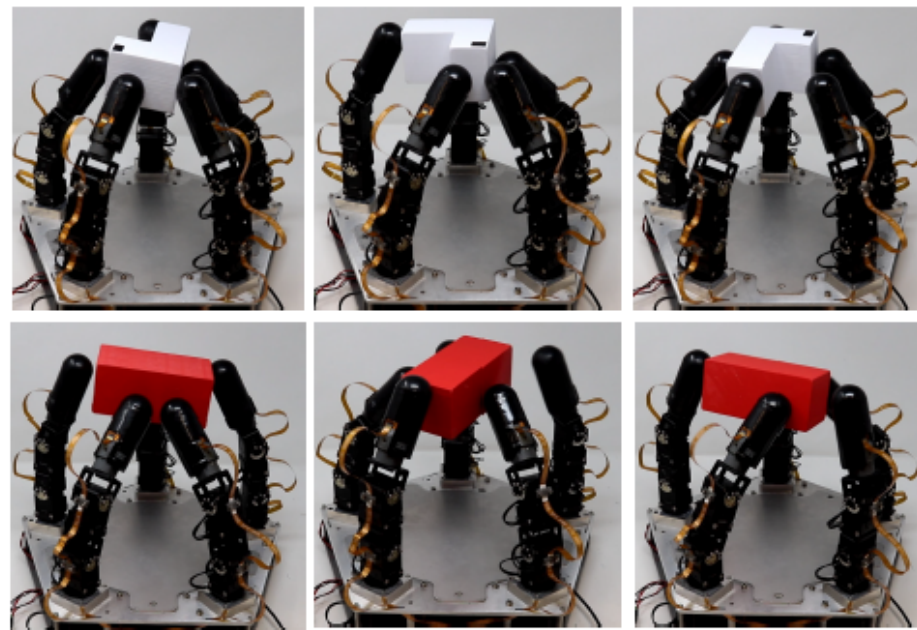
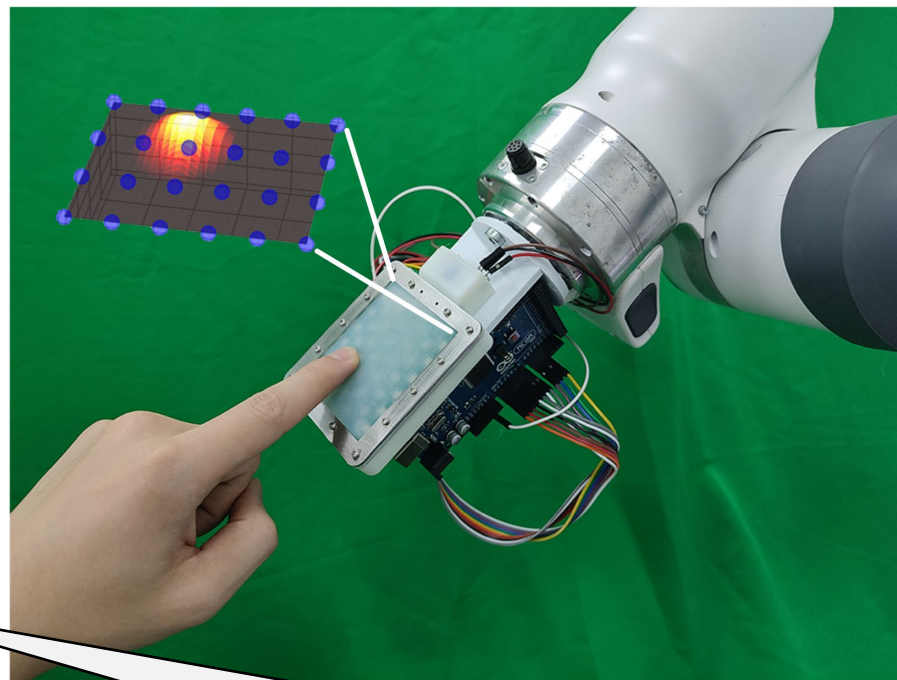


Fig. 1: Our method enables finger-gaiting manipulation of concave or elongated objects which require complex gaits. We demonstrate these gaits both in simulation and on a real robot hand (shown here), using only proprioceptive and tactile feedback intrinsic to the hand.

柔性触觉传感器



In case you need some tactile sensor for your design

A biomimetic tactile palm for robotic object manipulation

Ziwei Lei^{1,3}, Xutian Deng¹, Yi Wang¹, Xiaohui Xiao², Dong Han³, Fei Chen^{4,*}, Miao Li^{1,2,*}



2016/12/21

Why Tactile Intelligence Is the Future of Robotic Grasping - IEEE Spectrum

Why Tactile Intelligence Is the Future of Robotic Grasping

By Vincent Duchaine

Posted 18 Jul 2016 | 13:30 GMT

This is a guest post. The views expressed here are solely those of the author and do not represent positions of IEEE

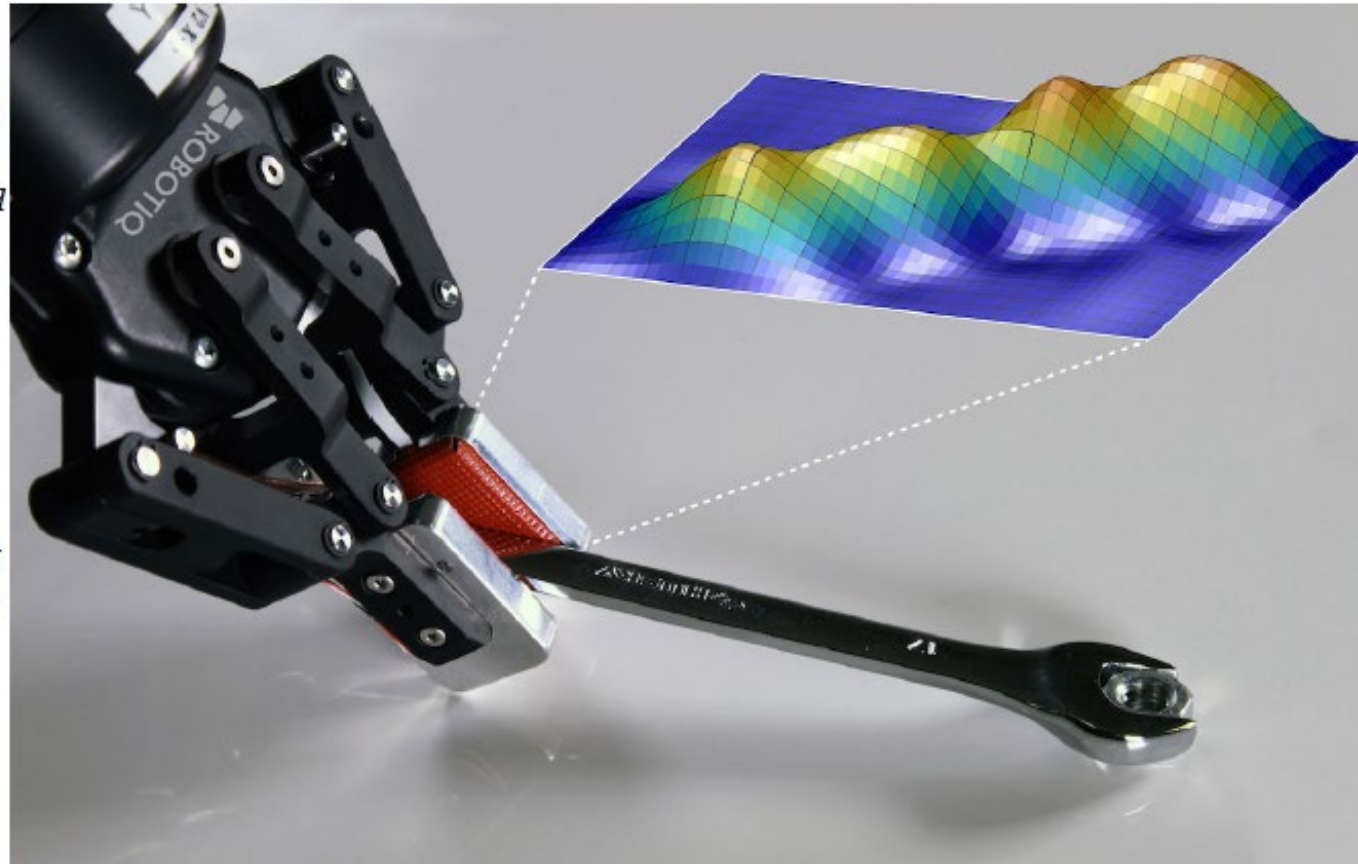


Image: CoRo Lab

Better tactile capabilities, not just vision, will let robots grasp any object.

A close-up photograph of a purple Northwestern University Mechanical Engineering marker. A robotic gripper, with white and black components, is holding the marker. The marker has the text "Northwestern University" and "Mechanical Engineering" printed on it. The word "Dexterity" is overlaid in large white font, and "Grand Challenges in Robotics" is overlaid in smaller white font above it.

Grand Challenges in Robotics

Dexterity

<https://robotics.northwestern.edu/grand-challenges/>



Hands in the Real World

Francesca Negrello^{1†}, Hannah S. Stuart^{2*†} and Manuel G. Catalano¹

¹ Soft Robotics for Human Cooperation and Rehabilitation, Italian Institute of Technology (IIT), Genova, Italy; ² Embodied Dexterity Group, Department of Mechanical Engineering, University of California at Berkeley, Berkeley, CA, United States

Robots face a rapidly expanding range of potential applications beyond controlled environments, from remote exploration and search-and-rescue to household assistance and agriculture. The focus of physical interaction is typically delegated to end-effectors—fixtures, grippers or hands—as these machines perform manual tasks. Yet, effective deployment of versatile robot hands in the real world is still limited to few examples, despite decades of dedicated research. In this paper we review hands that found application in the field, aiming to discuss open challenges with more articulated designs, discussing novel trends and perspectives. We hope to encourage swift development of capable robotic hands for long-term use in varied real world settings. The first part of the paper centers around progress in artificial hand design, identifying key functions for a variety of environments. The final part focuses on the overall trends in hand mechanics, sensors and control, and how performance and resiliency are qualified for real world deployment.

OPEN ACCESS

Keywords: hands, field robotics, dexterous manipulation, design, mechanisms, sensing, control, benchmarking



nature

GETTING A GRIP

An all-in-one method for creating programmable soft robots



Uncomfortable truths
Caltech confronts its past to build a more equitable future

Meet the ancestors
Genomes shed light on lineage of Bronze Age mummies

Counting the cost
Nature's salary survey reveals how COVID has affected the workplace

11 NOV 2021 10:00 AM GMT+0
nature.com

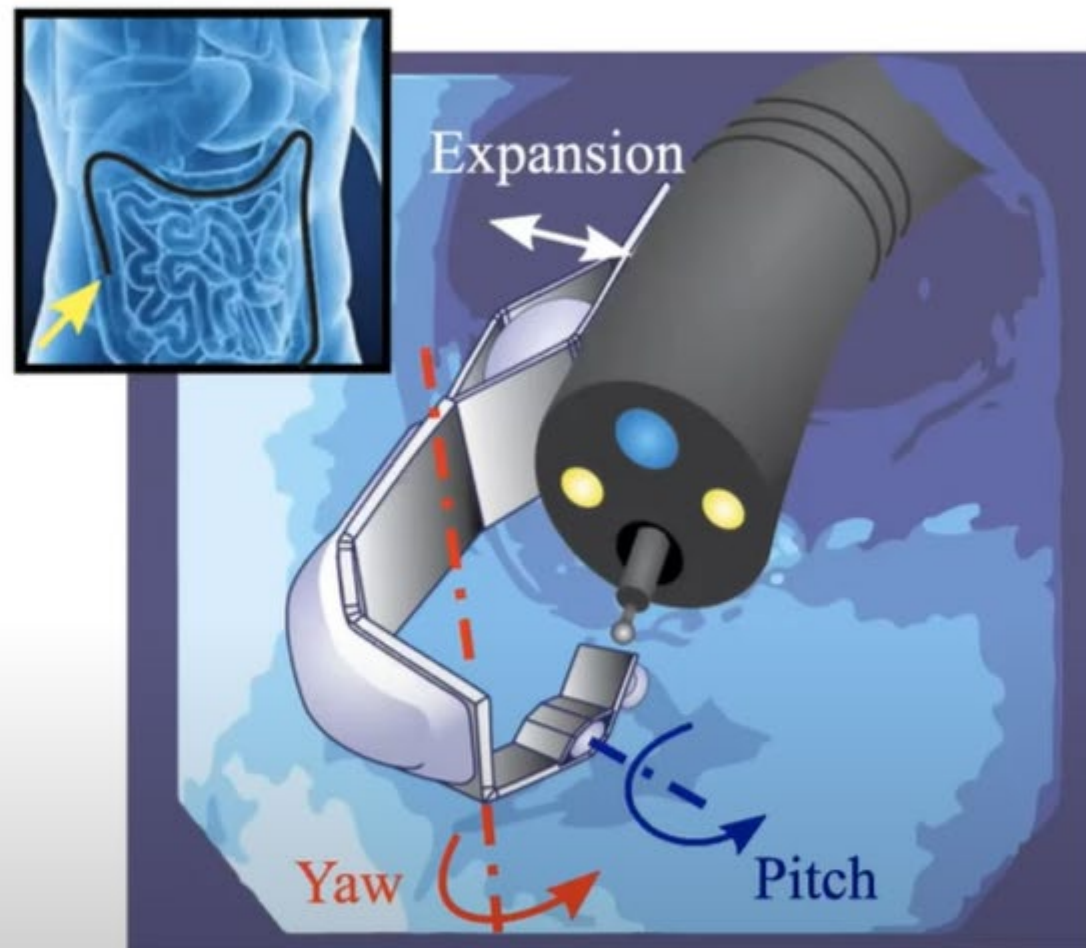
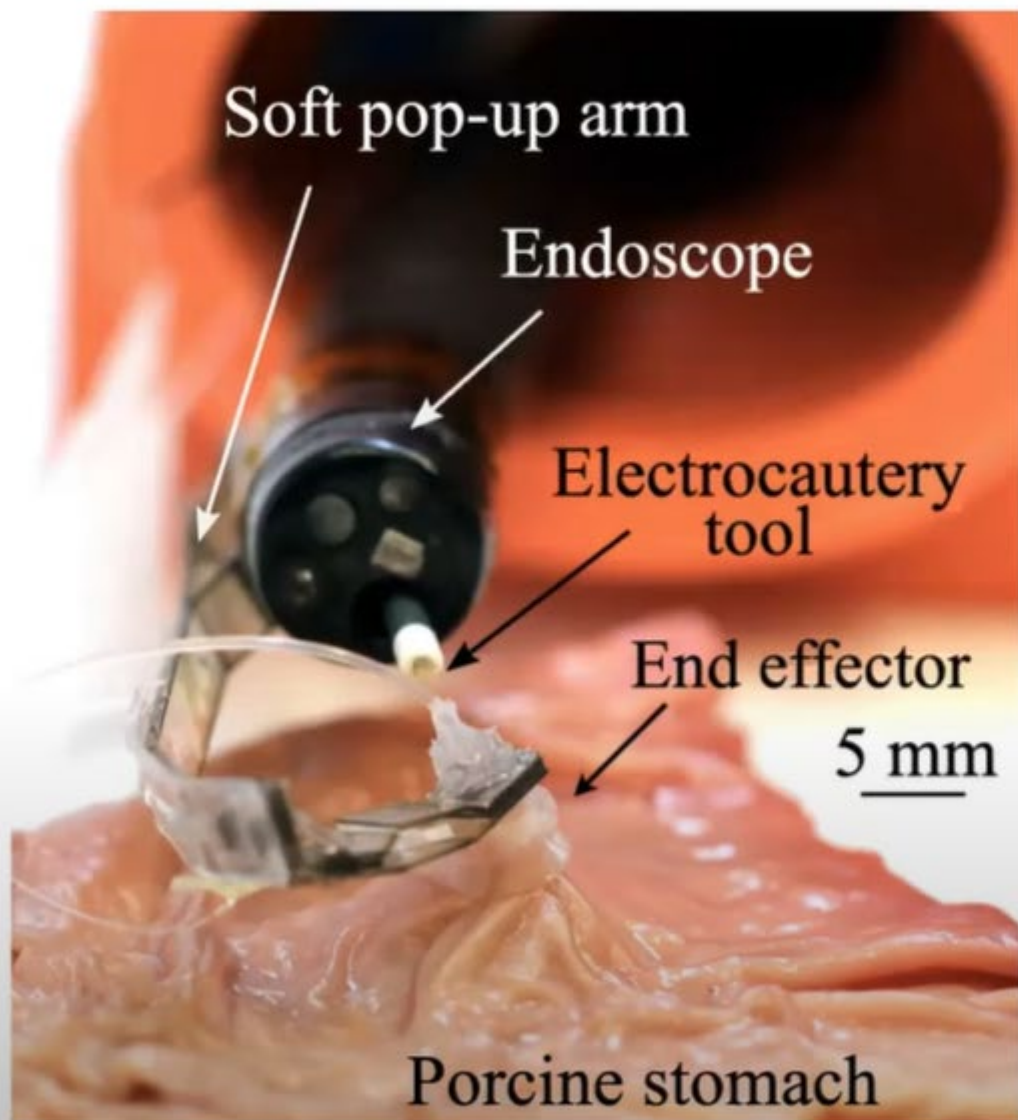




用吹气球的方式

五句精彩台词

Soft foldable robots for MIS

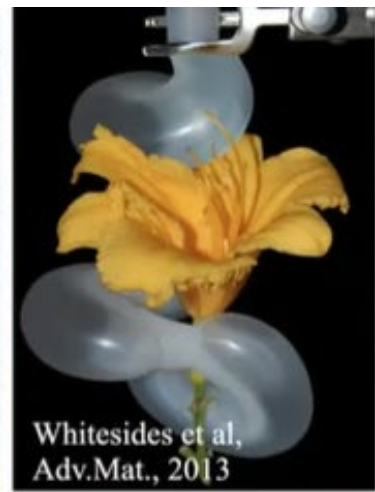




Laschi et al., Bioinspir. Biomim., 2012



Wood et al., Soft Robotics, 2016



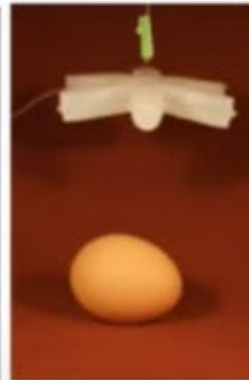
Whitesides et al, Adv.Mat., 2013



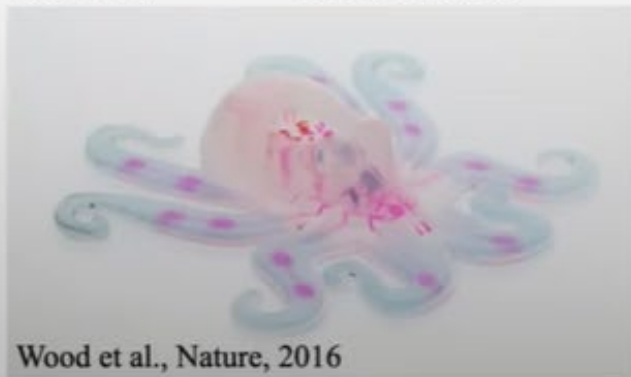
Shea et al., Adv. Mat., 2016



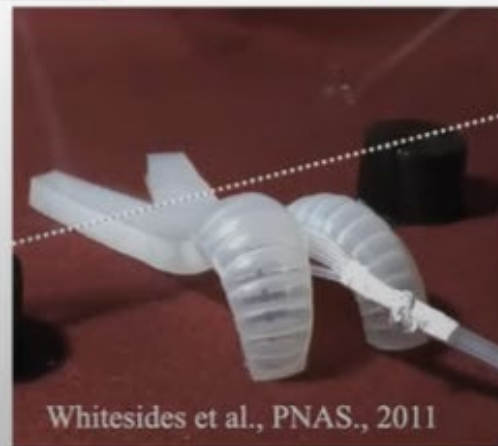
Soft Robotics Inc



Whitesides et al., Angew. Chem., 2010



Wood et al., Nature, 2016



Whitesides et al., PNAS., 2011

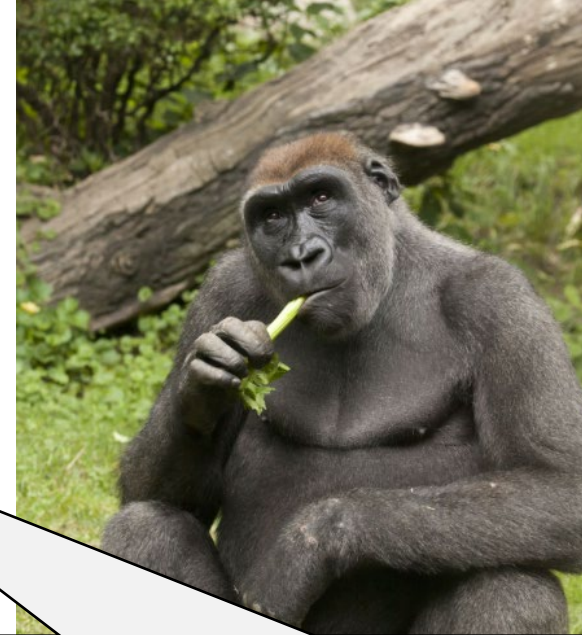


Seok et al., IEEE/ASME Trans. Mechatronics., 2013



Key components of a **new** hand

- **Body material**
- **Motion mechanism**
- **Actuator**
- **Sensor integrated**
- **Control algorithm**
- **Application scenarios**



Can be “new” in any sense



- Please select your project before next Thursday. (The hand project has the priority.)
- Try to use **5-10 slides** to explain your motivation, your brief concept, and why is interesting, etc ...
- Think outside the box!
- Simple, robust, easy to fabricate

